

Received 2 June 2025, accepted 19 June 2025, date of publication 27 June 2025, date of current version 9 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3583899

RESEARCH ARTICLE

Priority-Aware SDN Orchestration for Surgical IoMT: A Joint Optimization of Hit Ratio and Latency With Dynamic Resource Reallocation

YALDA SAFAEI¹, POURIA AREFIJAMAL², MAHDI SIAMAKI^{ID 2}, AND BARDIA SAFAEI^{ID 2}

¹School of Medicine, Hamadan University of Medical Sciences, Hamedan 14588-89694, Iran

²Department of Computer Engineering, Sharif University of Technology, Tehran 65178-38736, Iran

Corresponding author: Bardia Safaei (bardiasafaei@sharif.edu)

ABSTRACT Surgical Internet of Medical Things (IoMT) applications impose stringent Quality of Service (QoS) demands, requiring ultra-low latency and high reliability for critical data streams like control signals or high-definition video. Conventional resource orchestration methods often fall short, failing to guarantee performance for high-priority surgical tasks, particularly under dynamic network conditions, including varied wireless channel quality, and potential congestion. This paper introduces the Priority-Aware SDN Orchestration (PASO) framework, leveraging Software-Defined Networking (SDN) for centralized network visibility and control to address these critical challenges. PASO implements a novel joint optimization strategy focused primarily on maximizing the completion rate of critical surgical tasks that adhere strictly to their maximum delay constraints. A key distinguishing feature is its dynamic resource reallocation mechanism, which employs the preemption of lower-priority, non-medical traffic to proactively secure network resources and guarantee QoS for vital surgical flows, even during periods of high network load. A computationally efficient, priority-aware greedy algorithm operationalizes this strategy on the SDN controller. Comparative simulations against Random, Particle Swarm Optimization (PSO), Simulated Annealing (SA), and a Deep Reinforcement Learning (DRL) agent as baselines demonstrate PASO's substantial effectiveness. It significantly reduces end-to-end delay (e.g., up to 70% less than PSO under high load) and drastically improves the completion rate for critical surgical tasks (over 80% higher than PSO under high load), while also promoting balanced network utilization. PASO offers a robust solution for dependable, low-latency surgical IoMT operations through its targeted optimization and preemptive resource management capabilities.

INDEX TERMS Software-defined networking (SDN), internet of medical things (IoMT), surgical IoT, edge computing, task offloading, resource allocation, quality of service (QoS), latency optimization, hit ratio, preemptive scheduling, healthcare networks.

I. INTRODUCTION

The Internet of Medical Things (IoMT) is fundamentally reshaping healthcare delivery, forging an interconnected ecosystem of medical devices, sensors, and applications [1]. This paradigm enables enhanced patient care through advancements like remote monitoring, sophisticated diagnostics, and increasingly complex remote or assisted procedures

[2]. This domain's transformative potential and prevailing trends are well-documented [3], [4], signifying a major digital shift within the healthcare sector [5]. Underscoring this evolution, the global IoMT market is experiencing substantial growth, as depicted in Fig. 1, valued at US\$ 48.7 billion in 2022 and projected to surge to US\$ 370.9 billion by 2032, reflecting a Compound Annual Growth Rate (CAGR) of 23.15% [6]. Furthermore, the integration of IoMT with next-generation technologies like 6G promises even greater capabilities, particularly for massively connected healthcare scenarios [7].

The associate editor coordinating the review of this manuscript and approving it for publication was Hadi Tabatabaee Malazi^{ID}.

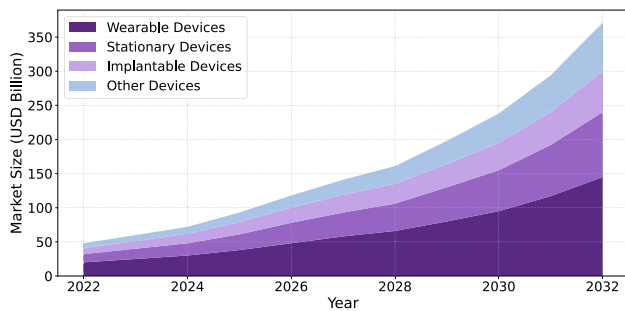


FIGURE 1. Projected global market growth for the IoMT, showing expected value from 2022 to 2032.

Among the diverse applications enabled by IoMT, surgical scenarios—spanning tele-surgery [8], robotic surgery assistance [9], real-time intraoperative diagnostics [10], and interactive guidance systems [11]—represent a particularly demanding frontier [12]. These applications are critically dependent on fulfilling stringent, non-negotiable Quality of Service (QoS) requirements. Specifically, they mandate ultra-low latency and extremely high reliability, often falling under the stringent umbrella of Ultra-Reliable Low-Latency Communication (URLLC) [13], [14], [15]. Ensuring the absolute integrity and punctual delivery of critical surgical data streams—such as haptic feedback, high-definition video, or real-time control commands—is paramount for patient safety and procedural success in these high-stakes environments.

Edge and fog computing architectures emerge as indispensable enablers for satisfying these exacting requirements [16], [17]. By locating computational resources geographically closer to IoMT devices compared to traditional centralized clouds, edge and fog platforms drastically reduce communication delays [18]. Task offloading further empowers resource-constrained medical devices to leverage these proximal computational capabilities for intensive operations, thereby optimizing both system responsiveness and device energy efficiency [19], [20]. Consequently, the effective allocation of resources and sophisticated scheduling of tasks across the multi-tiered computing infrastructure (spanning edge, fog, and potentially cloud) become pivotal for maximizing system performance [21], [22].

However, orchestrating these distributed resources specifically for the dynamic and demanding nature of surgical IoMT presents significant challenges. Network conditions can fluctuate due to factors such as wireless channel variability—an aspect further explored in our evaluations—and potential device mobility. Moreover, surgical procedures often generate unpredictable, bursty traffic patterns where critical data streams must invariably take precedence over less urgent data, such as routine system logs or non-essential background updates. Many contemporary task offloading and scheduling algorithms [23], [24], even sophisticated approaches employing techniques like deep reinforcement learning [25], frequently optimize for average performance

metrics (like average delay or energy) or lack the essential priority-awareness demanded by surgical contexts. Strategies relying on simple fairness models or assuming altruistic device behavior [26] prove inadequate when confronted with life-critical tasks demanding absolute priority and guaranteed timely completion within strict deadlines [27]. This highlights a critical gap: the need for frameworks capable of dynamic resource management with explicit prioritization and robust QoS guarantees for high-stakes surgical data, especially under network congestion or resource contention conditions.

Software-Defined Networking (SDN) presents a compelling architectural solution to address this complex orchestration challenge [28]. Its logically centralized control plane furnishes the requisite network-wide visibility and programmability to manage traffic flows dynamically, enforce granular QoS policies, and adaptively allocate resources based on real-time application requirements and defined priorities [29]. The capabilities of SDN can be further augmented by integrating specialized communication protocols optimized for the unique characteristics of medical data transmission [30]. Building upon this potential, our work introduces a tailored orchestration framework designed explicitly for surgical IoMT environments.

Our core motivation stems from the unyielding requirement to guarantee every life-critical surgical task's successful and timely processing. Standard best-effort network management or optimization based solely on average performance is fundamentally insufficient. We propose the Priority-Aware SDN Orchestration (PASO) framework, which employs a joint optimization strategy meticulously designed for the surgical IoMT context. The *primary objective* is to maximize the throughput of completed critical surgical tasks ($\mathcal{F}_m$) that strictly adhere to their designated maximum delay bounds ($D_{\max}$). While efficient resource utilization is essential, the framework prioritizes meeting these critical deadlines above all else. This explicit prioritization is enforced not only through the optimization model but critically, also via a dynamic resource reallocation mechanism featuring the *preemption* of lower-priority, non-medical traffic ($\mathcal{F}_n$). When network links approach congestion, PASO proactively reclaims bandwidth from less critical flows, thereby ensuring that vital surgical data streams meet their stringent QoS requirements. This proactive, priority-driven, and preemptive resource management distinguishes our approach and is essential for achieving the reliability demanded by surgical applications. This paper makes the following key contributions:

- 1) **Joint Optimization Formulation for Surgical IoMT:** A novel optimization model explicitly prioritizing the maximization of critical surgical task ($\mathcal{F}_m$) completion under strict delay constraints ($D_{\max}$).
- 2) **Priority-Aware Greedy Orchestration Algorithm:** An efficient greedy algorithm for the SDN controller, operationalizing the prioritized objective with latency-aware path selection and admission control.

3) **Dynamic Resource Reallocation with Preemption:**

A core mechanism enabling PASO to proactively guarantee resources for critical flows by preempting lower-priority traffic during potential congestion, ensuring adherence to $D_{\max}$.

4) **SDN-Based Implementation Framework:** Contextualization within an SDN architecture, highlighting how centralized control enables the necessary visibility and dynamic management for robust surgical IoMT orchestration.

Our simulations demonstrate the effectiveness of PASO. Compared to baseline methods like Random allocation, Particle Swarm Optimization (PSO), Simulated Annealing (SA), and a Deep Reinforcement Learning (DRL) agent, PASO significantly reduces end-to-end delay for completed tasks (with average reductions up to 70% against PSO and 25% against SA under high load) and dramatically increases the completion rate (deadline success rate) for critical surgical tasks (achieving improvements of approximately 82% over PSO, 50% over SA, and 200% over Random under high load). PASO also ensures more balanced network utilization. Our approach distinctively targets the maximization of *completed critical task throughput* under strict deadlines, leveraging joint optimization and preemption as key enablers.

The remainder of this paper is organized as follows: Section III introduces the system model and problem formulation. Section IV outlines the proposed orchestration algorithm. Section V explains the evaluation metrics, experimental setup, and the configuration of baseline algorithms, including the DRL agent. Section VI presents and analyzes the simulation results. Finally, Section VII concludes the paper and outlines potential directions for future research.

II. RELATED STUDIES

The convergence of SDN, the IoMT, and edge computing has catalyzed significant advancements in healthcare networks, particularly optimizing task offloading, resource allocation, and ensuring QoS. While our proposed PASO framework is evaluated against Random, PSO, SA, and an implemented DRL agent as baselines, it is important to acknowledge the broader spectrum of alternative SDN and QoS-aware orchestration techniques. These techniques are not only pivotal in IoMT but also find parallels in cutting-edge, non-IoMT orchestration frameworks and SDN-based systems applied in various other crucial fields, thereby enriching the conversation on advanced network management. Our review, while primarily focusing on the IoMT context, draws insights from this wider landscape. The discussed approaches encompass a variety of methods, including but not limited to, machine learning-based predictive algorithms, game-theoretic resource allocation models, and other sophisticated heuristic or meta-heuristic strategies tailored for different aspects of IoMT and edge/fog computing. This section provides a brief but comprehensive overview of several existing studies that reflect this diverse landscape, particularly those integrating these technologies in critical delay-sensitive

healthcare infrastructures, thereby providing a backdrop for our contributions.

In the context of task offloading and resource allocation, Ren et al. [31] delve into latency optimization in mobile-edge computation offloading, proposing joint communication and computation resource allocation strategies to minimize system delay. In another research, Liu et al. [32] introduce a dynamic task offloading framework that addresses ultra-reliable low-latency requirements by leveraging Lyapunov optimization and matching theory. Furthermore, Al-Masri [33] presents an edge-based resource allocation optimization tailored for IoMT, emphasizing multiple criteria decision-making to balance data privacy, costs, and latency. Chai et al. [34] explore time-slotted task offloading and resource allocation in cloud-edge-end cooperative computing networks, aiming to minimize task execution costs. Lakhan et al. [35] introduce the Joint Task Offloading and Scheduling (JTOS) framework, utilizing fuzzy multi-criteria methods to optimize task sequencing and scheduling in IoT systems, thereby reducing processing time and communication delays. In a related vein, Motamedhashemi et al. [36] presented FUSION, a task management policy for fog networks employing fuzzy logic for multi-objective optimization, specifically targeting a high Guarantee Ratio (GR) and minimal makespan. Also focusing on task scheduling in fog networks, Motamedhashemi et al. [37] proposed DATA, a genetic algorithm that optimizes for throughput and deadline adherence using a two-population strategy.

Meanwhile, as it has been summarized in Table 1, SDN can play a vital role in reducing latency and enhancing QoS for IoMT applications, especially in time-sensitive scenarios like executing medical tasks and performing telemetry/remote surgery. SDN separates the control plane from the data plane, allowing centralized controllers to have a global view of the entire network. This enables real-time, dynamic decision-making—like instantly rerouting traffic if congestion is detected—ensuring faster delivery of medical data. For instance, during remote surgery, if a video feed or control signal experiences delay, the SDN controller can detect the bottleneck and immediately reroute traffic through a lower-latency path [38], [39], [40]. Furthermore, traditional routing is often static. SDN allows dynamic, policy-based routing, which adapts based on current network conditions (e.g., congestion, link failure). This issue minimizes packet delays and jitter, which is crucial for continuous streams like real-time patient monitoring or haptic feedback in robotic surgery.

In addition, SDN is able to prioritize certain types of traffic (e.g., video feeds, patient vitals, actuator commands) over others. This ensures latency-sensitive tasks get fast-lane treatment, improving reliability and responsiveness for telemetry surgeries. For example, heart rate data or robotic control signals can be tagged as *high priority* and get guaranteed bandwidth and minimal delay. SDN controllers continuously monitor network conditions (e.g., latency, throughput, link status). As a result, if the network

TABLE 1. Key SDN capabilities, corresponding IoMT benefits, and their impact on latency.

SDN Capability	Benefit to IoMT	Impact on Latency
Centralized Control	Real-time routing decisions	Lower latency
Dynamic Path Optimization	Avoids congested routes	Lower jitter/delay
Traffic Prioritization	Critical data gets priority	Faster task execution
Edge Integration	Local computation	Shorter RTT
Real-Time Monitoring	Rapid adaptation	Smoother performance
Fast Failover	Continuous operation	No lag spikes

starts degrading, SDN can reallocate resources or shift workloads to nearby edge nodes, reducing dependency on distant cloud servers and cutting round-trip times. SDN can also orchestrate tasks between the cloud, edge, and devices, choosing the optimal execution locations. Therefore, latency-critical tasks (e.g., motion feedback in robotic surgery) are executed at the edge, close to the data source, while SDN ensures the fastest data paths are used. In the event of a link failure or node crash, SDN provides fast failover, automatically rerouting traffic and reducing downtime and latency spikes, which is critical in surgical operations where milliseconds matter. Due to the above-mentioned aspects, several researchers have concentrated on integrating IoMT with SDN infrastructures to enhance QoS. The authors in [41] discuss the benefits of decoupling control software from hardware, facilitating efficient task offloading and resource allocation in IoT applications. Also, authors in [42] focus on optimizing response time in SDN-edge environments, proposing a unified approach that considers both networking and computing dimensions to meet the stringent QoS demands of IoT applications. Addressing the challenges in vehicular edge computing, another study introduces a multi-objective approach for joint task offloading and resource allocation in SDN-enabled environments. This approach balances latency and energy consumption, which is crucial for time-sensitive healthcare applications [43]. In another study, authors propose a prioritization-based delay-sensitive task offloading strategy in SDN-integrated mobile IoT networks, utilizing gravitational search algorithms for optimal fog node selection, resulting in significant reductions in delay and unassigned tasks [44].

In the realm of healthcare, an investigation into re-scheduling IoT services in edge networks has been conducted, which offers insights into optimizing resource utilization and maintaining QoS, which are critical for healthcare networks dealing with fluctuating workloads [45]. Edge computing is not just about offloading but also about intelligent decision-making at the edge. Integrating AI/ML models at the edge enables real-time predictions, anomaly detection, and context-aware resource allocation. A few studies present adaptive frameworks that learn from usage patterns to optimize resource use dynamically [40], [46]. For instance, Siyadatzaheh et al. [47] proposed ReLIEF, a reinforcement learning-based primary-backup task assignment strategy for fault-tolerant fog computing,

aiming to enhance reliability for real-time IoT tasks. Surgical IoT systems, in particular, demand ultra-low latency and deterministic behavior. For example, Huang et al. and Luo et al. showed that pre-configured SDN policies, combined with intelligent task migration and bandwidth reservation, are effective for these critical environments [48], [49]. Furthermore, the authors in [50] propose a joint task offloading and resource allocation strategy for mobile edge computing-enabled medical vehicular networks to reduce latency and improve resource utilization. Li et al. [51] present a resource allocation and data offloading strategy for edge-computing-assisted intelligent telemedicine systems, introducing a contract mechanism to incentivize edge servers and enhance system utility. Furthermore, the dynamic nature of IoT services necessitates adaptive scheduling mechanisms. Moreover, several optimization-based approaches have been employed to enhance task offloading efficiency, focusing on minimizing energy consumption and delay. Simulated annealing techniques have been widely applied in this regard [52], [53], [54], [55], demonstrating their effectiveness in optimizing task scheduling and offloading strategies in mobile edge computing environments. In parallel, particle swarm optimization (PSO) methods have also been utilized to achieve low-latency task offloading and efficient resource allocation [56], [57], [58], offering robust solutions for edge-cloud and industrial IoT applications.

In addition to these approaches focusing on system optimization and resource management, other lines of research investigate fundamental aspects of network reliability and performance characterization. For example, Safaei et al. [59] introduce a novel metric to evaluate the capability of routing protocols in mobile IoMT environments to establish and maintain network connections, employing Markov chain analysis for its calculation. This highlights the ongoing efforts to develop more comprehensive metrics for understanding network behavior beyond traditional ones.

In the realm of advanced network optimization, DRL has emerged as a pivotal technique for addressing complex challenges in dynamic environments. Recent investigations underscore its effectiveness in both task offloading and packet routing. For instance, Ren et al. propose a DRL-based distributed task offloading framework designed for edge-cloud environments, which aims to achieve efficient and near-optimal offloading decisions by enabling intelligent agents to learn adaptive strategies [60]. Complementing this, in the context of packet routing, Li et al. present a DRL-based joint routing and capacity optimization approach specifically tailored for aerial and terrestrial hybrid wireless networks, focusing on dynamically minimizing end-to-end packet transmission delay [61]. These studies highlight DRL's capability to learn and execute sophisticated control policies, thereby significantly enhancing performance metrics in modern networked systems.

As Table 2 illustrates, each category presents distinct advantages and limitations in the context of highly dynamic

TABLE 2. Comparison of optimization and learning methods in dynamic surgical IoMT environments.

Feature/Characteristic	Metaheuristic	Greedy	Reinforcement Learning (RL)	Machine Learning (ML)
Robustness	Generally robust [62]	Often robust [63]	Sensitive to changes [64]	Vulnerable to attacks/data shift [65], [66]
Time to Convergence	Moderate [67]	Fast [68]	Slow, inefficient [67], [67]	Varies by model/training [69]
Dynamic Environments	Adaptable via re-run [70]	Poor adaptation [71]	Strong adaptation [72]	Needs retraining [73]
Data Dependency	Low [74]	Low [68]	High [72]	High [69]
Parameter Tuning	High [75]	Low [68]	High (hyperparameters/reward) [73]	High (hyperparameters) [65]
Convergence Guarantee	Probabilistic [70]	Deterministic [68]	Probabilistic [76]	Data/algorithm-dependent [77]
Computational Cost	Moderate–High [78]	Low [68]	High (train), Moderate (inference) [72]	High (train), Low (inference) [69]

environments like surgical IoT. Specifically, Reinforcement Learning (RL) methods, while capable of learning from interactions and adapting to dynamic conditions, often suffer from a high time to convergence [67], [83]. This prolonged training period, coupled with their inherent sample inefficiency, makes them less suitable for real-time critical decision-making where immediate adaptation is required. Furthermore, the learned policies can be sensitive to environmental changes if not sufficiently explored during training. On the other hand, Machine Learning (ML) methods, particularly supervised learning approaches, excel at pattern recognition and prediction given sufficient labeled data. However, their robustness is a significant concern in the face of unforeseen data shifts or adversarial attacks, which are plausible in complex IoMT environments [84]. Moreover, ML models often struggle to directly optimize sequential decision-making problems in dynamic settings without extensive re-training, and their “black-box” nature can hinder interpretability, crucial for medical applications. They fundamentally operate on learning from existing data rather than actively exploring and optimizing within a continuously changing environment. Greedy methods, while offering fast execution, inherently make locally optimal choices. This characteristic means they cannot guarantee global optimality and are often ill-equipped to handle the complex, non-linear interdependencies and long-term consistency required in surgical IoT scenarios [85]. Their lack of foresight makes them unsuitable for problems where future states or cumulative effects are critical. Given these limitations, particularly the critical need for consistency over time in a rapidly evolving surgical IoT environment, coupled with the slow convergence of RL, the robustness issues of ML, and the sub-optimality of greedy approaches, we propose a novel metaheuristic method. Our approach leverages the strengths of metaheuristics in achieving near-optimal solutions within a reasonable time and adapting to complex, dynamic landscapes, offering a promising solution for maintaining crucial consistency in surgical IoMT operations.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section formally defines the system architecture, operational parameters, decision variables, and the mathematical optimization problem underlying our proposed Priority-Aware SDN Orchestration (PASO) framework for surgical IoMT environments. We consider a network environment orchestrated by a central SDN controller, connecting

various IoMT devices to geographically distributed edge computing resources capable of task processing and data caching. A graphical representation of the system architecture is depicted in Fig. 2. The flowchart in Fig. 3 demonstrates the decision-making logic for handling high and low priority tasks in the controller. It highlights delay-aware routing and resource allocation to ensure efficient transmission.

A. SYSTEM ARCHITECTURE AND COMPONENTS

The physical infrastructure comprises a set of network nodes $\mathcal{V}$, which includes IoMT source devices, edge computing nodes, and intermediate routers or switches. These nodes are interconnected by a set of logical communication links $\mathcal{L}$, managed by the SDN controller. Each link $l \in \mathcal{L}$ is characterized by its data transmission capacity C_l (e.g., in bits per second) and its associated communication delay d_l , which encompasses propagation and transmission times.

A crucial infrastructure component is the set of edge computing nodes $\mathcal{E} \subset \mathcal{V}$. These nodes are strategically located closer to the IoMT devices to facilitate low-latency processing. Each edge node $e \in \mathcal{E}$ possesses a finite cache storage capacity B_e (e.g., in bits) and introduces a processing delay p_e when executing an offloaded task, reflecting its computational power and the task’s complexity.

The communication model integrates IoMT devices within the surgical environment, which connect via wireless (e.g., Wi-Fi, BLE) or wired links to local Access Points (APs). These APs act as gateways to a high-speed, low-latency backhaul network (e.g., fiber-optic), ensuring reliable data transmission to edge nodes or, if necessary, to the central cloud. The entire network fabric, including APs, links, and edge nodes, operates under the centralized control of the SDN controller. This controller maintains a global view of the network topology and resource status, enabling dynamic routing, bandwidth allocation, and caching decisions. At the data link layer, we assume a priority-aware Medium Access Control (MAC) protocol that grants preferential channel access to high-priority medical data streams, minimizing contention delays for critical traffic. For the performance and QoS focus of this work, we assume robust end-to-end encryption (e.g., TLS/DTLS) for data confidentiality and integrity in transit. However, the detailed design and analysis of security mechanisms specific to the PASO framework, including aspects beyond in-transit protection such as endpoint security or SDN controller access control, are identified as key future research directions.

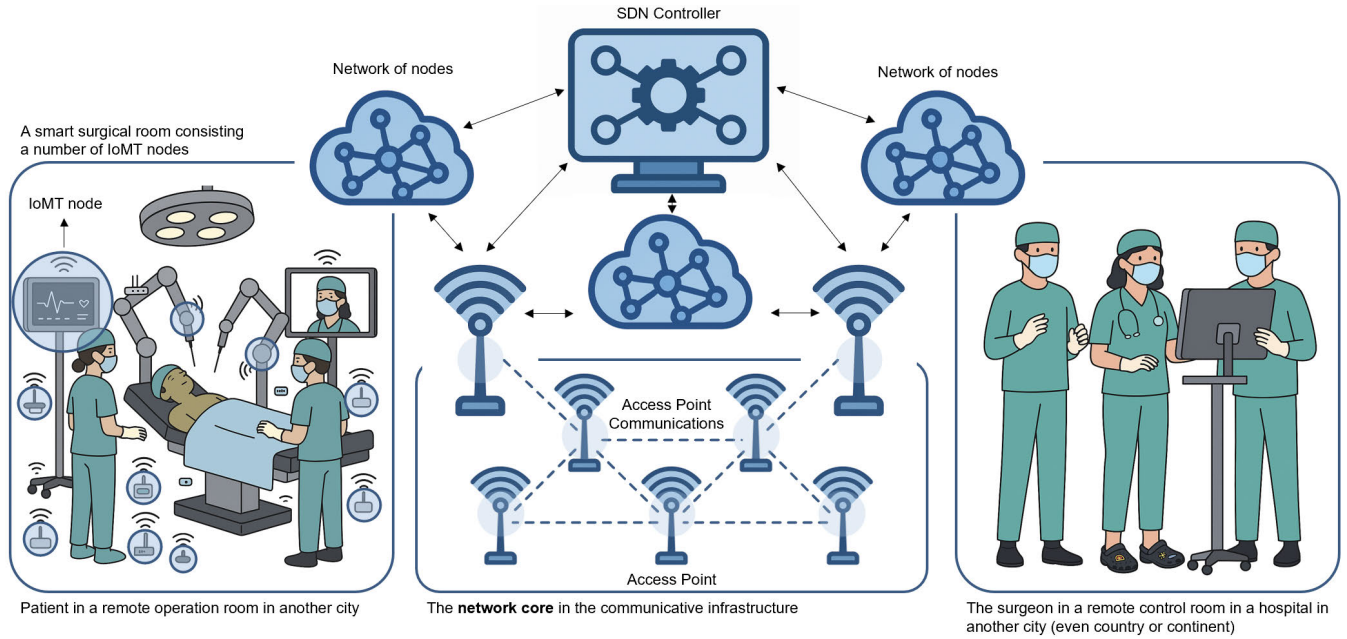


FIGURE 2. System architecture for the surgical IoMT environment, showing IoMT devices, the central SDN controller, and communication links.

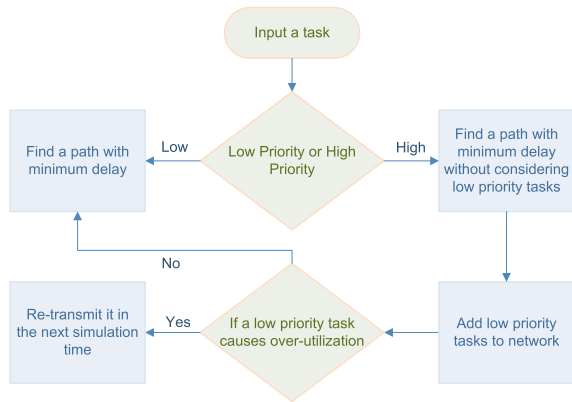


FIGURE 3. Flowchart illustrating the task prioritization and path selection process based on delay minimization.

B. TASKS, FLOWS, AND QOS REQUIREMENTS

Over a given time horizon, the system handles a set of tasks or data flows $\mathcal{F}$, generated by IoMT devices. These tasks are categorized into two distinct subsets:

- $\mathcal{F}_m \subset \mathcal{F}$: High-priority medical tasks, such as surgical video, control signals, or critical patient telemetry, which demand stringent QoS guarantees.
- $\mathcal{F}_n \subset \mathcal{F}$: Lower-priority non-medical tasks, like routine logs or background data synchronization ($\mathcal{F} = \mathcal{F}_m \cup \mathcal{F}_n$, $\mathcal{F}_m \cap \mathcal{F}_n = \emptyset$).

Each task $r \in \mathcal{F}$, originating from a source node $\text{src}(r) \in \mathcal{V}$, is characterized by its data size s_r (relevant for caching), its required network bandwidth b_r , and its intended processing destination $\text{dst}(r)$, which is typically an edge node $e \in \mathcal{E}$

or potentially the cloud. The energy consumed by the source device to transmit task r for offloading is denoted by E_r .

Specific QoS requirements must be met to ensure patient safety and procedural efficacy in surgical applications. A critical parameter is the maximum acceptable end-to-end delay $D_{\max}$ for any admitted medical task $r \in \mathcal{F}_m$. Additionally, a minimum average hit ratio $H_{\min}$ is targeted specifically for admitted medical tasks to promote efficient utilization of edge resources for critical data. For battery-powered IoMT devices, an implicit constraint might be the device's maximum energy budget $E_{\max}$ for transmissions over time.

C. DECISION VARIABLES

The SDN controller dynamically manages the network by making the following key decisions for each task $r \in \mathcal{F}$:

- **Admission Control** ($z_r \in \{0, 1\}$): Determines if task r is admitted and successfully processed ($z_r = 1$) or rejected/dropped ($z_r = 0$).
- **Routing Decision** ($y_{rl} \in \{0, 1\}$): Indicates if link $l \in \mathcal{L}$ is used in the path for transmitting an admitted task r ($y_{rl} = 1$ if used, 0 otherwise).

These variables collectively define the orchestration strategy employed by the controller.

D. OPTIMIZATION PROBLEM FORMULATION

Our primary objective is to maximize the number of successfully processed tasks, while strongly prioritizing critical medical tasks. This is formulated to maximize a weighted sum of completed tasks, subject to network and QoS constraints.

1) OBJECTIVE FUNCTION

We aim to maximize the weighted sum of successfully admitted and processed tasks, where medical tasks receive full weight and non-medical tasks receive a fractional weight α ($0 < \alpha < 1$) to enforce prioritization:

$$\text{Maximize : } \sum_{r \in \mathcal{F}_m} z_r + \alpha \sum_{r \in \mathcal{F}_n} z_r \quad (1)$$

Maximizing this objective drives the system towards admitting as many medical tasks as possible, followed by non-medical tasks, within the limits imposed by the constraints.

2) CONSTRAINTS

The maximization of the objective function (1) is governed by several critical constraints that ensure operational feasibility and adherence to the stringent requirements of the surgical IoMT environment. Firstly, Quality of Service (QoS) for critical medical tasks must be guaranteed. The end-to-end delay experienced by any admitted medical task ($r \in \mathcal{F}_m$ with $z_r = 1$), encompassing the cumulative delay across its network path ($\sum y_{rl} d_l$) and the processing time at the assigned edge node ($p_{e(r)}$), must strictly remain below the maximum threshold $D_{\max}$:

$$\left(\sum_{l \in \mathcal{L}} y_{rl} d_l + p_{e(r)} \right) z_r \leq D_{\max} \quad \forall r \in \mathcal{F}_m. \quad (2)$$

Secondly, the system must operate within its finite resource limitations. Network congestion is managed by ensuring that the aggregate bandwidth consumed by all active tasks ($r \in \mathcal{F}$) traversing any link l does not surpass the link's inherent capacity C_l . Significantly, this constraint incorporates the dynamic preemption mechanism, allowing the available capacity on a link to be effectively increased by reclaiming bandwidth from lower-priority non-medical tasks ($r' \in \mathcal{F}_n$) that are preempted to accommodate critical flows:

$$\sum_{\substack{r \in \mathcal{F} \\ \text{active on } l}} y_{rl} b_r \leq C_l + \sum_{\substack{r' \in \mathcal{F}_n \\ \text{being preempted on } l}} y_{r'l} b_{r'} \quad \forall l \in \mathcal{L}. \quad (3)$$

Thirdly, valid network routing paths must be established, which the flow conservation constraint ensures. For any task r and any intermediate network node v (i.e., a node that is neither the source $\text{src}(r)$ nor the destination $\text{dst}(r)$), the total incoming flow must equal the total outgoing flow:

$$\sum_{l \in \text{in}(v)} y_{rl} = \sum_{l \in \text{out}(v)} y_{rl} \quad \forall r \in \mathcal{F}, \forall v \notin \{\text{src}(r), \text{dst}(r)\}. \quad (4)$$

Finally, all the decision variables introduced – admission control (z_r), and routing (y_{rl}) – are inherently binary, meaning they can only take values of 0 or 1:

$$z_r, y_{rl} \in \{0, 1\} \quad \forall r \in \mathcal{F}, \forall l \in \mathcal{L}. \quad (5)$$

Together, these constraints define the operational boundaries within which the optimization objective (1) must be achieved, capturing the complex interplay between QoS demands, resource availability, and network logic in the surgical IoMT system.

IV. PROPOSED GREEDY ALGORITHM WITH PREEMPTION

Addressing the computational complexity inherent in optimally solving the Integer Linear Programming problem formulated in Section III within the dynamic constraints of real-time network control necessitates a practical, efficient heuristic. We propose the Priority-Aware SDN Orchestration (PASO) algorithm, a computationally tractable, greedy heuristic designed to operate on the central SDN controller. PASO makes localized, sequential decisions for each arriving task, aiming to approximate the optimal resource allocation strategy defined by our objective function (1). Its core principles are the strict prioritization of critical medical traffic ($\mathcal{F}_m$) and dynamic resource management, incorporating the preemption of lower-priority non-medical tasks ($\mathcal{F}_n$) when network congestion threatens the Quality of Service (QoS) of vital surgical data flows.

The orchestration logic is detailed in Algorithm 1. The process begins by sorting the set of incoming tasks $\mathcal{F}$ based on priority, ensuring that all medical tasks ($\mathcal{F}_m$) are considered before any non-medical tasks ($\mathcal{F}_n$) (Line 1). This initial sorting step directly implements the prioritization central to our objective. The algorithm then iterates through each task r in this prioritized sequence (Line 2).

For each task r , an initial feasibility check is performed by invoking the 'FIND_DECISION' procedure (Algorithm 2, called in Line 3). This procedure (detailed in Algorithm 2) acts as a preliminary filter. It compares an estimated offloading delay D_r (potentially based on a typical path or target edge node) against the stringent maximum allowed delay $D_{\max}$ for the task type and verifies if the originating device has sufficient energy E_r for the transmission ('Device_Has_Energy', Line 1). If either the estimated delay exceeds the threshold or the device lacks energy, offloading is deemed infeasible ('false' returned, Line 4), and the algorithm defaults to instructing the device to compute the task locally ('Compute_Locally', Line 26), assuming local processing capability exists. The system state is then updated accordingly (Line 27).

If the task r passes the initial feasibility check ('can_offload' is true, Line 4), PASO attempts to find a suitable network path for offloading. It employs a standard shortest path algorithm, such as Dijkstra's ('Dijkstra_Find_Path', Line 5), configured to find a route from the task's source node $\text{src}(r)$ to a suitable edge node $e \in \mathcal{E}$ that minimizes the end-to-end latency, considering current link delays d_l and edge processing delays p_e .

If a potential path is identified (Line 6 evaluates to true), the algorithm proceeds to the critical step of verifying resource availability along this path. First, a flag 'preempt_allowed' is set to true only if the task r is a high-priority medical task ('Is_Medical(r)' returns true, Line 7), granting it the potential to trigger preemption. The algorithm then meticulously checks each link l composing the selected path (Line 8). For every link, the 'Calculate_Utilization' function (Line 9) estimates the resulting link utilization if task r , requiring bandwidth b_r , were to be admitted. Crucially, this

Algorithm 1 Priority-Aware Task Orchestration with Preemption (PASO)

Input: Tasks Set ($\mathcal{F}$), Links Set ($\mathcal{L}$), Edge Nodes Set ($\mathcal{E}$), Devices Set ($\mathcal{D}$), Maximum Delay ($D_{\max}$), Energy Limits

Output: Optimized task offloading and routing decisions

/ Priority-Aware Task Orchestration */*

```

1 Sort tasks in  $\mathcal{F}$  by priority (medical tasks  $\mathcal{F}_m$  first);
2 for Each task  $r$  in  $\mathcal{F}$  do
3    $can\_offload \leftarrow \text{FIND\_DECISION}(r, D_r, D_{\max}, E_r)$ ;
4   /* Check feasibility */
5   if  $can\_offload$  then
6      $path \leftarrow \text{Dijkstra\_Find\_Path}(r, \mathcal{L}, \mathcal{E}, \text{minimize\_latency})$ ;
7     /* Find best path */
8     if  $path$  exists then
9        $preempt\_allowed \leftarrow \text{Is\_Medical}(r)$ ; /* Only
10      medical tasks can preempt
11       $path\_feasible \leftarrow \text{true}$ ; for Each link  $l$  in  $path$  do
12        /* Check resource availability */
13        if  $\text{Calculate\_Utilization}(l, r, preempt\_allowed) > 0.90$  then
14           $path\_feasible \leftarrow \text{false}$ ; break;
15          /* Congested link
16        end
17      end
18      if  $path\_feasible$  then
19         $\text{Admit\_and\_Route}(r, path, preempt\_allowed)$ ; /* Allocate
20        resources
21         $\text{Update\_System\_State}(r, path)$ ;
22      end
23      else
24         $\text{Defer\_Task}(r)$ ; /* No feasible path due
25        to congestion
26      end
27    end
28    else
29       $\text{Defer\_Task}(r)$ ; /* No path found
30    end
31  end
32 else
33    /* Cannot offload (failed feasibility
34    check) */
35     $\text{Compute\_Locally}(r)$ ; /* Fallback to local
36    execution
37     $\text{Update\_System\_State}(r, \text{local})$ ;
38  end
39 end
40 return Optimized task offloading and routing decisions;
```

calculation is priority-aware: if ‘preempt_allowed’ is true, the function considers the available bandwidth plus any bandwidth that could potentially be reclaimed by preempting currently active, lower-priority non-medical tasks ($\mathcal{F}_n$) on that specific link l . If admitting task r would cause the anticipated utilization on any link l to exceed a predefined operational threshold (e.g., 90%, chosen to maintain a safety margin and network stability), even after accounting for potential preemption, that link is considered congested for this task. In such a case, the path is marked as infeasible (‘path_feasible’ set to false, Line 13), and the checking loop for this path terminates (‘break’).

If the entire path is successfully checked without exceeding the utilization threshold on any link (‘path_feasible’ remains true, Line 13), the task r is admitted. The ‘Admit_and_Route’ function (Line 14) is invoked to allocate the necessary

Algorithm 2 Task Offloading Feasibility Check

Input: Task (r), Estimated Offload Delay (D_r), Maximum Delay ($D_{\max}$), Device Energy Requirement (E_r)

Output: Boolean (true if offloading feasible)

/ Task Offloading Feasibility Check */*

```

1 if ( $D_r \leq D_{\max}$ ) and Device_Has_Energy( $E_r$ ) then
2   return true; /* Offloading is feasible
3 end
4 else
5   return false; /* Offloading not feasible
6 end
7 return Boolean;
```

resources formally. This involves reserving the bandwidth b_r on each path link and updating cache status if applicable. Importantly, if preemption was required to meet the resource needs (as determined during the ‘Calculate_Utilization’ check when ‘preempt_allowed’ was true), this function also executes the necessary preemption actions, instructing the network elements to deprioritize or drop specific non-medical flows on the congested links. Following successful resource allocation, the overall system state (link loads, cache usage) is updated (‘Update_System_State’, Line 15).

However, if the path verification failed due to anticipated congestion on one or more links (Line 13 evaluates to false), the task r cannot be accommodated on this path at this time and is deferred (‘Defer_Task’, Line 18). Similarly, if the initial ‘Dijkstra_Find_Path’ call failed to find any viable path to an edge node (Line 6 evaluates to false), the task is also deferred (Line 22). Deferred tasks might be reconsidered later or dropped depending on the system’s policy.

This iterative, priority-driven greedy approach allows PASO to dynamically manage network resources, making real-time decisions that strongly favor the successful and timely completion of critical surgical tasks. Preemption is leveraged as a key mechanism to guarantee QoS under potential network load.

The computational complexity of the PASO algorithm (Algorithm 1) is primarily determined by the task sorting step and the pathfinding within the loop. Sorting $N = |\mathcal{F}|$ tasks takes $O(N \log N)$ time. For each task, the dominant operation is typically the Dijkstra’s algorithm for pathfinding, which has a complexity of $O(L + V \log V)$ or $O((V + L) \log V)$ for a network with $V = |\mathcal{V}|$ nodes and $L = |\mathcal{L}|$ links, using a binary heap. The subsequent link verification loop runs for at most $V - 1$ links. Thus, the overall complexity is approximately $O(N \log N + N \cdot (L + V \log V))$. While this is polynomial, for very large networks or extremely high task arrival rates, the controller’s processing capacity needs to be sufficient to maintain real-time responsiveness.

V. EVALUATION METHODOLOGY

We employ a set of standard quantitative metrics to evaluate the performance and effectiveness of our proposed priority-aware SDN orchestration framework. These metrics are chosen to directly assess the framework’s ability to meet the objectives outlined in Section III, particularly the

prioritization of medical tasks ($\mathcal{F}_m$) while adhering to system constraints. We will compare the performance of PASO against relevant baseline algorithms under various simulated network conditions and task loads, as detailed in Section VI. The key metrics are defined as follows:

A. TOTAL DELAY (D_{TOTAL})

To comprehensively evaluate the end-to-end latency performance of the system, we define the Total Delay as the cumulative sum of the completion delays experienced by all successfully processed medical tasks. For each task r , the completion delay d_r^{comp} is the time elapsed from the task's arrival to its successful completion at an edge or fog node. The Total Delay across all completed medical tasks is formulated as follows:

$$D_{\text{total}} = \sum_{r \in \mathcal{F}_m} d_r^{\text{comp}} \quad (6)$$

A lower D_{total} indicates better system responsiveness and faster processing of critical medical tasks, which is crucial for maintaining Quality of Service (QoS) guarantees in time-sensitive IoMT environments.

B. HIT RATIO (H)

In the IoMT setting, the Hit Ratio evaluates the system's effectiveness in ensuring that offloaded medical tasks meet their deadlines after traversing the network. Since all tasks are transmitted to edge or fog nodes (i.e., no local execution occurs), we define the Hit Ratio as the proportion of tasks that are completed within their deadline constraints [86]:

$$H = \frac{\sum_{r \in \mathcal{F}} \mathbb{I}(d_r^{\text{comp}} \leq D_r^{\text{deadline}})}{|\mathcal{F}|} \quad (7)$$

where d_r^{comp} represents the completion time of task r (measured from arrival to successful completion at an edge or fog node), D_r^{deadline} is the assigned deadline for task r , and $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 if the condition is satisfied (i.e., the task meets its deadline) and 0 otherwise. A higher H value indicates better system performance, demonstrating that a greater fraction of IoMT tasks are being completed on time. We particularly focus on the Hit Ratio for high-priority medical tasks (H_m), which serves as a direct measure of the framework's ability to provide reliable and timely data processing essential for medical and surgical applications.

C. LINK UTILIZATION (U_l)

This metric reflects how efficiently network bandwidth resources are utilized and helps identify potential bottlenecks or imbalances across the network. In our simulation, the instantaneous utilization of a link l during a task transfer was calculated using the following expression, implemented in Python [87]:

$$U_l(t) = 100 \times \left(1 - \frac{\text{AvailableCapacity}_l(t)}{C_l} \right) \quad (8)$$

In addition to average link utilization, we introduce the 90th Percentile Deviation from Mean Utilization (Δ_{90}) to better capture the fairness and balance of network resource usage. This metric quantifies how much, on average, the utilization of the most utilized 90% of links deviates from the overall mean utilization across all links. It is formally defined as:

$$\Delta_{90} = \frac{1}{|\mathcal{L}_{90}|} \sum_{l \in \mathcal{L}_{90}} |U_l - \mu_U| \quad (9)$$

where $\mathcal{L}_{90}$ denotes the set containing the 90% of links with the highest average utilizations, U_l is the average utilization of link l , and μ_U is the mean utilization across all links. Lower values of Δ_{90} indicate that most links have utilizations close to the average, implying a more balanced and efficient network operation. Higher values suggest uneven load distribution and potential bottlenecks in specific links. This approach allows us to assess not only the overall bandwidth usage but also the fairness and efficiency of the load distribution, which is critical for maintaining low latency and avoiding network congestion, especially in highly dynamic IoMT environments.

D. DYNAMIC CONDITIONS: WIRELESS NOISE SIMULATION SETUP

To evaluate the robustness of the proposed PASO framework under more realistic and dynamic wireless channel conditions, we conducted simulations incorporating wireless noise. This allows for an assessment of how varying signal quality impacts key performance indicators such as hit ratio and end-to-end delay. The simulation of noise and its effects are based on established wireless communication principles and models. The core idea is that the Signal-to-Noise Ratio (SNR) at the receiver dictates the Bit Error Rate (BER), which in turn affects the Packet Error Rate (PER). A higher PER leads to more retransmissions, increasing delay and potentially reducing effective throughput [88], [89].

1) NOISE GENERATION AND IMPACT

In our simulations, wireless channel quality is primarily modulated by adjusting the SNR. We consider two primary noise models conceptually, although the simulations presented in Section VI-E directly vary SNR to observe performance changes:

Elevated AWGN Floor: This model assumes a constant Additive White Gaussian Noise (AWGN) power (P_{N1}) across the channel [90]. The SNR is then determined by the received signal power (P_{rx}) relative to this noise floor, calculated as:

$$\text{SNR}_{dB} = P_{rx,dBm} - P_{N1,dBm} \quad (10)$$

Here, $P_{rx,dBm}$ is the received signal power in dBm, and $P_{N1,dBm}$ is the noise power in dBm. Fluctuations in performance arise from changes in received signal strength or if this noise floor itself is varied.

AWGN + Bursty Interference: This model considers a baseline AWGN ($P_{N,base}$) plus intermittent, higher-power interference (P_I) that is active with a certain probability or

duty cycle (ρ) [91]. When interference is active, the total noise power increases significantly. The total noise power in linear units ($P_{N,Total_B_linear}$) is the sum of the baseline noise and the interference power:

$$P_{N,Total_B_linear} = P_{N,base_linear} + P_{I_linear} \quad (11)$$

This leads to a temporary drop in SNR, calculated as $SNR_B = P_{rx_linear} / P_{N,Total_B_linear}$, and consequently a higher PER [91]. This models common scenarios in shared frequency bands, such as the ISM band, where devices like Bluetooth or microwave ovens can cause transient interference [92]. The average PER, and consequently system performance, fluctuates based on the presence and characteristics of this interference. The average PER is a weighted sum of the PER in each state:

$$PER_{avg} = (1 - \rho) \cdot PER(SNR_A) + \rho \cdot PER(SNR_B) \quad (12)$$

where SNR_A is the SNR when only baseline noise is present [88].

In both noise models, the fundamental principle is that different noise components are added together (in their linear power representations) to determine the total effective noise power. This total noise power, when combined with the received signal power, yields the SNR. The calculated SNR then directly influences the BER and PER, ultimately impacting overall system performance metrics like delay and throughput.

For the results presented in Section VI-E, we directly vary SNR levels from 10 dB to 25 dB. Lower SNR values represent noisier, more challenging channel conditions, while higher SNR values represent clearer channels. This variation in SNR impacts the underlying BER and PER. Although not explicitly simulating the full MAC-layer retransmission protocol for each packet in the main simulation loop for computational efficiency, the performance degradation (increased delay, reduced hit ratio) shown in the results reflects the expected impact of higher PER values that would result from lower SNR conditions. The relationship between SNR, BER, PER, and effective data rate is well-documented [88], [89].

2) KEY PARAMETERS FOR NOISE SIMULATION

The parameters relevant to understanding the wireless environment and noise impact are summarized in Table 3. These parameters are drawn from typical IoT and wireless communication scenarios and relevant standards [93].

The simulation study investigates the performance of PASO and baseline algorithms across a range of SNR levels (10 dB to 25 dB). The Python script, used to generate the comparative plots under varying noise (SNR) conditions, models the degradation in hit ratio and the increase in sum delays as SNR decreases. Specifically, a baseline performance at a nominal high SNR is established, and then scaling factors, proportional to the deviation from this nominal SNR, are applied to simulate the impact of worsening channel conditions. For instance, as SNR drops

TABLE 3. Key parameters for wireless noise simulation setup.

Wireless Channel & General Parameters	
IoT Device Transmit Power (P_{tx})	60 mW (≈ 17.78 dBm) [84]
Received Signal Power (P_{rx})	-70 dBm
Wireless Channel Bandwidth (B)	20 MHz [84]
Packet Length (L , Approx.)	3600 bits [82]
Conceptual Noise Model 1: Elevated AWGN Floor	
Noise Power (P_{N1})	-81 dBm [73], [85]
Conceptual Noise Model 2: AWGN + Bursty Interference	
Baseline AWGN Power ($P_{N,base}$)	-95 dBm [82]
Interference Power (P_I , active)	-70 dBm [83], [86]
Interference Duty Cycle (ρ)	0.2 (20%) [70]
Simulated SNR Range for Performance Evaluation	
Simulated SNR Levels	10, 15, 20, 25 dB

from 25 dB towards 10 dB, leading to a reduction in hit ratios and an increase in delays, with slight random variations to simulate stochastic channel effects. This approach, while abstracting the detailed physical layer calculations for each packet, provides a clear trend of performance sensitivity to wireless noise across the compared orchestration methods.

By analyzing these metrics across different scenarios and comparing PASO with baseline approaches, we can provide a comprehensive assessment of its performance in jointly optimizing task throughput, delay, and resource utilization for priority-aware surgical IoMT orchestration.

E. DEEP REINFORCEMENT LEARNING (DRL) BASELINE

To provide a comprehensive comparison against contemporary machine learning-based scheduling approaches, and in response to the need for benchmarking against state-of-the-art techniques, we implemented a DRL agent for the task offloading decision process. The DRL agent learns a policy to decide whether to drop, execute locally, or offload an incoming IoT task to a fog node, aiming to optimize a predefined reward signal. We utilized the Stable Baselines3 library [96] for this implementation. The converged DRL model, after training (performance detailed in Section VI-B), serves as an advanced baseline for performance comparison against PASO and other heuristic methods.

1) DRL ENVIRONMENT FORMULATION

The task scheduling problem is modeled as a Markov Decision Process (MDP), where the DRL agent interacts with a custom-simulated IoT scheduling environment.

a: STATE SPACE

The state observed by the agent at each step is represented as a 6-dimensional vector of normalized values, capturing essential task and network characteristics relevant to the scheduling decision. These include the normalized task size, obtained by dividing the task size by 500.0 KB, which corresponds to typical IoMT data payloads [97]; the normalized task deadline, calculated as the task deadline divided by 20.0 ms, reflecting the requirements of latency-sensitive IoMT applications [98]; and the normalized battery capacity of the task's source IoT device, computed by dividing the

battery capacity by 6000.0 units, indicative of standard battery levels in IoT devices. In addition, the vector contains the task priority, where a value of 1.0 denotes high priority and 0.0 otherwise; the average link utilization across the network; and the normalized queue length of the least busy fog node, obtained by dividing the queue length by 10.0 tasks. All state values are normalized and clipped to the range [0, 1]. The normalization constants (MAX_TASK_SIZE, MAX_DEADLINE, MAX_BATTERY) are defined within our DRL environment to reflect typical task properties encountered in IoMT scenarios [97], [98].

b: ACTION SPACE

The DRL agent in the IoT task scheduling environment operates within a discrete action space comprising three possible actions. Action 0 corresponds to dropping the task, where the task is simply discarded. Action 1 represents local execution, in which the agent attempts to process the task directly on the IoT device. Action 2 involves offloading the task to the least busy fog node for remote processing.

c: REWARD FUNCTION

The reward function is crafted to steer the agent towards making optimal scheduling decisions by providing feedback based on the outcomes of each action. If the agent selects Action 0 (drop task), it receives an immediate negative reward of -1.0 . For Action 1 (local execution), if the task's local execution energy E_{local} exceeds the available device battery B_{device} , the reward is -0.5 , indicating battery insufficiency. If the task can be executed energy-wise but the current time $T_{current}$ plus the local execution duration D_{local} exceeds the task deadline $T_{deadline}$, the agent receives a reward of -0.8 , signaling a deadline miss. In the case where both battery and deadline constraints are satisfied, a positive reward of $+1.0$ is granted for successful local execution. For Action 2 (offloading to a fog node), the agent first checks if the offloading transmission energy $E_{offload}$ surpasses the device's battery B_{device} , which yields a reward of -0.5 if true. If the energy constraint is satisfied but the total offloading delay $D_{offload}$ added to $T_{current}$ exceeds $T_{deadline}$, the agent again receives -0.8 for missing the deadline. If neither constraint is violated, the reward is $+1.0$, indicating successful offloading. For any undefined or unexpected cases, a default reward of 0.0 is applied. This structured reward scheme—featuring $+1.0$ for success, -0.5 for battery insufficiency, -0.8 for deadline violations, and -1.0 for dropped tasks—is intended to encourage the agent to prioritize successful task completion within energy and timing constraints, thereby promoting efficient resource utilization. This approach aligns with common practices in DRL-based resource management strategies [99]. Additionally, the environment returns a `hit` boolean variable indicating whether the task was successfully processed (i.e., not dropped and meeting battery and deadline conditions in local or offloaded modes).

Algorithm 3 Deep Q-Learning for IoMT Task Scheduling

Input: Environment (Env), Q-Network (Q), Target Network (Q'), Replay Buffer ($\mathcal{D}$), Learning Rate (α), Discount Factor (γ), Exploration Rate (ϵ), Target Update Frequency (C), Batch Size (B), Total Timesteps (T_{total})

Output: Trained DQN model for IoMT task scheduling

```

/* Initialization */
1 Initialize  $Q$  with random weights  $\theta$ ; Initialize target network  $Q'$  with  $\theta' \leftarrow \theta$ ; Initialize empty replay buffer  $\mathcal{D} \leftarrow \emptyset$ ; Set timestep  $t \leftarrow 0$ ; Set initial state  $s \leftarrow Env.reset()$ ;

/* Main Training Loop */
2 for  $t \leftarrow 1$  to  $T_{total}$  do
    /*  $\epsilon$ -greedy action selection */
    3 if  $rand() < \epsilon$  then
    4      $a \leftarrow$  random action; // Explore randomly
    5 else
    6      $a \leftarrow \arg \max_{a'} Q(s, a'; \theta)$ ; // Exploit learned policy
    7 end
    8  $r, s', done \leftarrow Env.step(a)$ ; // Apply action and observe transition
    9 Store transition  $(s, a, r, s', done)$  in buffer  $\mathcal{D}$ ;

    /* Experience Replay and Learning */
    10 if  $|\mathcal{D}| \geq B$  and  $t \geq \text{warm-up threshold}$  then
    11     Sample mini-batch  $(s_j, a_j, r_j, s'_j, done_j)$  from  $\mathcal{D}$ ; for each sampled transition do
    12         if  $done_j$  then
    13              $y_j \leftarrow r_j$ ;
    14         else
    15              $y_j \leftarrow r_j + \gamma \max_{a''} Q'(s'_j, a''; \theta')$ ;
    16         end
    17         Compute loss:  $L(\theta) \leftarrow \frac{1}{B} \sum_{j=1}^B (y_j - Q(s_j, a_j; \theta))^2$ ;
    18         Update  $Q$ :  $\theta \leftarrow \theta - \alpha \nabla_{\theta} L(\theta)$ ;
    19     end
    20 end

    /* Target Network Update */
    21 if  $t \bmod C = 0$  then
    22      $\theta' \leftarrow \theta$ ; // Synchronize target network
    23 end
    24  $s \leftarrow s'$ ; // Update current state
    25 if  $done$  then
    26      $s \leftarrow Env.reset()$ ; // Reset environment on episode end
    27 end
28 return Trained Q-network  $Q$ ;

```

2) DRL ALGORITHM AND TRAINING PROTOCOL

We employed the Deep Q-Network (DQN) algorithm [100], a value-based DRL algorithm suitable for discrete action spaces. The DQN agent learns an action-value function (Q-function) that estimates the expected return for taking each action in a given state. The policy is then derived by selecting the action with the highest Q-value. The training process is detailed in Algorithm 3.

The DRL agent was trained for 100,000 timesteps. Key hyperparameters for the DQN agent, are summarized in Table 4. These parameters were chosen based on common practices in DRL research and some empirical tuning, with values like learning rate and buffer size being standard starting points [96], [101]. An Adam optimizer was used for training the neural network.

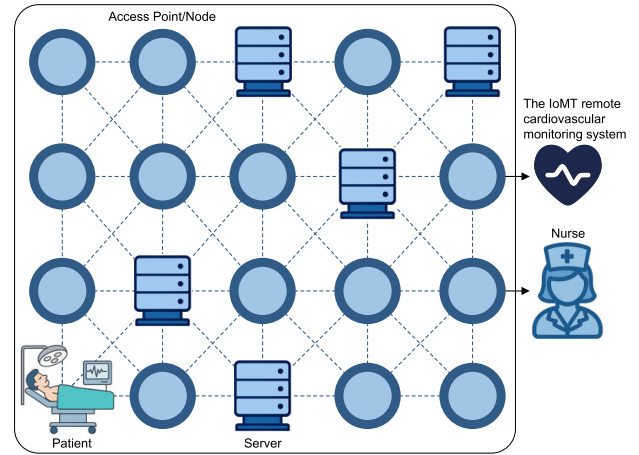
TABLE 4. DQN hyperparameters for IoT task scheduler.

Parameter	Value
DRL Algorithm	DQN
Policy Network	MlpPolicy (Multi-Layer Perceptron)
Learning Rate	5e-4 [92]
Replay Buffer Size	100,000 samples
Learning Starts	5,000 steps
Batch Size	128
Discount Factor (γ)	0.99
Target Update Interval (C)	10,000 steps
Exploration Initial Epsilon	1.0
Exploration Final Epsilon	0.05
Exploration Fraction	0.1 (Portion of training to anneal epsilon)
Total Timesteps	100,000

Algorithm 3 outlines the training procedure for the DQN agent designed for IoMT task scheduling. The process commences with an initialization phase (Line 1). Here, the primary Q-network (Q) is initialized with random weights θ , and its counterpart, the target network (Q'), is initialized with an identical set of weights $\theta' \leftarrow \theta$. A replay buffer $\mathcal{D}$ is created and left empty, ready to store the agent's experiences. The simulation timestep t is set to zero, and the environment Env is reset to provide the initial state s for the agent.

The core of the algorithm is the main training loop (Lines 2-27), which executes for a predefined number of T_{total} timesteps. In each timestep t (Line 2), the agent employs an ϵ -greedy strategy to select an action a (Lines 3-7). With a probability of ϵ , a random action is chosen to encourage exploration of the action space (Line 4). Otherwise, with probability $1 - \epsilon$, the agent exploits its current learned policy by selecting the action a that yields the maximum Q-value $Q(s, a'; \theta)$ for the current state s (Line 6). Once an action is selected, it is applied to the environment via $Env.step(a)$, which returns the immediate reward r , the subsequent state s' , and a boolean flag *done* indicating whether the current episode has terminated (Line 8). This experience tuple $(s, a, r, s', done)$ is then stored in the replay buffer $\mathcal{D}$ (Line 9).

The learning phase, leveraging experience replay, begins when the replay buffer $\mathcal{D}$ has accumulated at least B (Batch Size) transitions and the training has progressed beyond an initial warm-up period (Line 10). A mini-batch of B transitions $(s_j, a_j, r_j, s'_j, done_j)$ is randomly sampled from $\mathcal{D}$ (Line 11). For each transition within this mini-batch (Line 12), a target Q-value y_j is computed. If the episode corresponding to the transition had terminated (*done_j* is true), y_j is simply the observed reward r_j (Line 13). If the episode had not terminated, y_j is calculated using the Bellman equation: $r_j + \gamma \max_{a''} Q'(s'_j, a''; \theta')$, where γ is the discount factor and Q' is the target network (Line 15). The use of the target network Q' for calculating these target values enhances training stability. A loss function $L(\theta)$, typically the mean squared error, is then computed between these target values y_j and the Q-values $Q(s_j, a_j; \theta)$ predicted by the main Q-network (Line 17). The weights θ of the main Q-network Q are subsequently updated by performing a gradient descent step with a learning rate α to minimize this loss $L(\theta)$ (Line 18).

**FIGURE 4.** Representative scale-free network topology (Barabási-Albert) used for simulating the surgical IoMT environment.

To further stabilize learning, the target network Q' is updated periodically. Every C timesteps (Target Update Frequency), the weights θ' of the target network are synchronized with the weights θ of the main Q-network: $\theta' \leftarrow \theta$ (Lines 20-22). After the learning update, the current state s is advanced to s' (Line 23). If the *done* flag indicates the end of an episode, the environment is reset, and s is reinitialized to the starting state of the new episode (Lines 24-26). Upon completion of T_{total} timesteps, the algorithm returns the trained Q-network Q , which encapsulates the learned scheduling policy.

VI. SIMULATION RESULTS AND DISCUSSION

In this section, we evaluate the performance of the proposed Priority-Aware SDN Orchestration (PASO) framework. First, we present the training performance of the DRL agent, which serves as one of our baselines. Subsequently, we compare PASO against four baseline methods: random task allocation, PSO-based allocation, SA-based allocation, and the trained DRL agent. The evaluation focuses on key performance metrics introduced in Section V, across diverse network conditions simulated through different topological deployments of fog nodes and IoMT devices. The heuristic and metaheuristic baseline methods include simulated annealing-based task offloading approaches [52], [53], [54], [55], where task allocation decisions are optimized via iterative probabilistic searches, and particle swarm optimization-based techniques [56], [57], [58], which leverage swarm intelligence to minimize latency and energy consumption during task scheduling. The DRL baseline provides a comparison against a learning-based approach trained to optimize task scheduling decisions.

A. SIMULATION SETUP

Key simulation parameters defining the network configuration, device capabilities, and task characteristics are summarized in Table 5. The simulations were conducted

TABLE 5. Simulation environment and parameters.

Network Configuration	
Simulation Area	500 × 500 m ²
Topology Type	Barabási–Albert (scale-free)
Access Points (APs)	15
Connected APs	Fiber Channel [93]
Fog Nodes	5
SDN Controller Algorithm	OpenFlow [94]
Wireless Standard	IEEE 802.11 [95]
Inter-AP Bandwidth	10 Gbps [96]
Edge Link Bandwidth	100 Mbps [97]
Noise Power	−100 dB [97]
Device Specifications	
Fog Node CPU Frequency	2.9–4.2 GHz [88]
IoT CPU Frequency	16–84 MHz [88]
IoT Transmission Power	60 mW [98]
Offloading Threshold	20%
Task Parameters	
Number of Tasks	100–1000
Average Task Size	450 KB [88]
Computation Demand	1500–2500 Megacycles [97]
Deadline Range	1–5 ms [89]
Priority Distribution	Uniform
Simulation Setup	
Processor	Intel Core i7 (12th Gen)
Memory	16 GB RAM
GPU	NVIDIA GTX 1650
Storage	1 TB SSD
Programming Language	Python

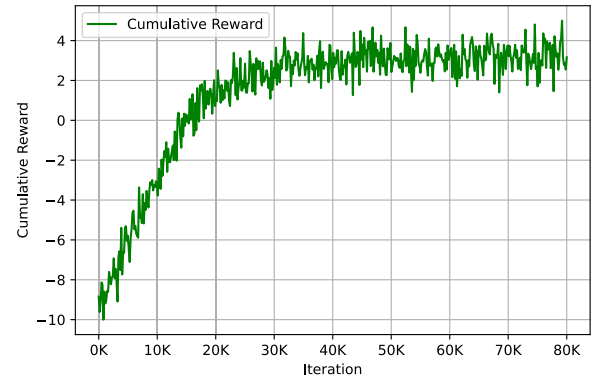
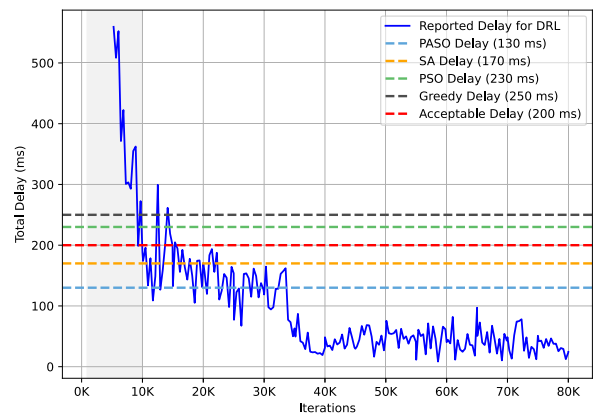
within a 500 × 500 m² area using a scale-free Barabási–Albert network topology [102] to model the access network structure. The environment included 15 Access Points (APs) and 5 Fog Nodes, shown in Fig. 4. To assess robustness, results are presented across four different random placements (Topologies 1–4) of fog nodes and IoMT devices. The simulations were executed on a Windows machine with specifications detailed earlier. The implementation was done in Python.

B. DRL AGENT CONVERGENCE AND TASK DELAY TRENDS

Before its use as a baseline, the DRL agent (DQN) was trained for 100,000 timesteps as described in Section V-E. The learning efficacy of the DRL agent is illustrated by its training progress. Fig. 5 shows the reward obtained by the agent over training timesteps, typically averaged over a sliding window or per episode. The convergence or plateauing of the reward curve, visible after approximately 80,000 timesteps, indicates that the DRL agent has learned an effective scheduling policy, maximizing its cumulative reward.

Fig. 6 illustrates the total delay experienced by tasks across training iterations for the DRL agent, in comparison with four methods: PASO, SA, PSO, and Greedy. As established in the literature, for latency-sensitive applications such as video streaming and remote robotics—including surgical monitoring—end-to-end delays should ideally remain below 200 ms to ensure acceptable QoS [109].

While the DRL agent demonstrates a gradual reduction in delay as training progresses—ultimately achieving optimized latency after sufficient exposure to diverse task

**FIGURE 5. DRL agent reward per training iteration/episode, demonstrating learning convergence.****FIGURE 6. Comparison of total delay across training iterations for DRL, PASO, SA, PSO, and Greedy methods.**

patterns—its performance in the early training phases is suboptimal, particularly when the network is under high load and dominated by lower-priority tasks. In contrast, the PASO framework maintains consistently lower delay from the outset, even under dynamic load conditions and heterogeneous task priorities. The SA, PSO, and Greedy methods exhibit progressively higher average delays, reaching approximately 170 ms, 230 ms, and 250 ms, respectively.

Although the DRL agent eventually outperforms most baselines in later training iterations, PASO consistently achieves lower delay throughout the entire training period. This is especially crucial in scenarios involving surgical tasks, where minimizing latency is vital. As the DRL agent's policy improves, the total delay for successfully completed tasks stabilizes at a lower value, reflecting its increasing ability to manage latency effectively once an optimal routing policy is learned.

C. PERFORMANCE EVALUATION OF HEURISTIC-BASED METHODS

We compare PASO against Random, PSO, and SA focusing on delay, hit ratio, and network utilization.

1) DELAY AND LATENESS PERFORMANCE

Low latency is paramount in surgical IoMT. Fig. 7 presents the total end-to-end delay averaged across all successfully completed tasks as the number of tasks increases from 100 to 1000, shown across the four simulated topologies.

a: OBSERVATIONS

Across all topologies, the total delay generally increases with the task load for all algorithms, as expected, due to increased resource contention. However, the performance characteristics diverge significantly. PASO demonstrates a controlled increase in delay. Even under the highest load (1000 tasks), PASO maintains a significantly lower delay compared to PSO and SA. For instance, in Topology 1 at 1000 tasks, PASO's delay (approx. 206ms) is roughly 65% lower than PSO's (approx. 588ms) and 15% lower than SA's (approx. 243ms). Similar trends hold across other topologies, with PASO consistently achieving delays 55-70% lower than PSO and 10-25% lower than SA under heavy load. This superior performance stems from PASO's ability to proactively manage resources via priority-aware path selection and preemption, securing low-latency paths for critical tasks even when the network is busy.

Both metaheuristic approaches, PSO and SA, show a much steeper increase in delay, particularly beyond 500 tasks. Their delays escalate rapidly, suggesting they struggle to find efficient global solutions under high contention, leading to significant queuing delays and suboptimal routing choices. This implies a higher likelihood of violating the stringent delay threshold $D_{\max}$ for critical medical tasks, resulting in increased lateness (D_{total}). Random exhibits the lowest apparent total delay. However, this is highly misleading. As shown later in the hit ratio analysis (Fig. 8), the Random approach achieves this low average delay by simply dropping a large fraction of tasks, especially complex or high-priority ones, that might induce higher delays. It processes only the "easiest" tasks, rendering its low average delay figure unrepresentative of effective system performance for critical applications.

b: ANALYSIS

PASO's effectiveness in delay management, especially under load, highlights the benefit of its integrated SDN control, priority queuing, and preemption mechanisms. By ensuring critical tasks ($\mathcal{F}_m$) are processed first and non-critical tasks ($\mathcal{F}_n$) can be preempted, PASO minimizes deadline violations for surgical data, directly addressing the core requirement of surgical IoMT. The high delays seen with PSO and SA indicate their potential unsuitability for real-time critical systems without significant modifications for priority awareness and congestion handling.

2) HIT RATIO PERFORMANCE

The hit ratio, representing the percentage of submitted tasks successfully processed at an edge node, is crucial for

evaluating both task completion success and edge resource utilization efficiency, especially concerning the constraint $H_{\min}$ for medical tasks. Fig. 8 compares the hit ratios for both average tasks (solid lines) and high-priority surgical tasks (dashed lines) across the different task loads and topologies.

a: OBSERVATIONS

PASO consistently achieves the highest hit ratio for both average and surgical tasks across all task loads and topologies. Notably, the hit ratio for surgical tasks remains very high (typically above 60-75% even at 1000 tasks), significantly outperforming all baselines. For example, in Topology 1 at 1000 tasks, PASO's surgical hit ratio (approx. 60%) is roughly 82% higher than PSO's (approx. 33%), 50% higher than SA's (approx. 40%), and about 200% higher than Random's (approx. 20%). Furthermore, the gap between PASO's average and surgical hit ratio is relatively small, indicating that its prioritization mechanism successfully favors critical tasks without excessively penalizing non-critical ones when resources allow.

PSO and SA show moderate initial hit ratios that degrade significantly as the task load increases. Their hit ratios for surgical tasks are considerably lower than PASO's, especially under high load (dropping to 30-40% at 1000 tasks). This indicates their struggle to guarantee resources for high-priority tasks when the system is congested, failing to meet the demands of reliable surgical IoMT. Random performs the poorest, with hit ratios plummeting rapidly as the number of tasks increases. Its hit ratio for surgical tasks falls below 20-25% under high load, demonstrating a complete lack of effective prioritization or resource management.

b: ANALYSIS

PASO's superior hit ratio, particularly for surgical tasks, is a direct consequence of its design. The algorithm explicitly prioritizes $\mathcal{F}_m$ tasks (Algorithm 1, Line 1) and employs preemption (Algorithm 1, Lines 7-14) to reclaim resources from $\mathcal{F}_n$ tasks when necessary. This ensures that critical surgical data streams are successfully offloaded and processed, maximizing the primary objective (Equation 1) and fulfilling the reliability requirements. The poor performance of Random highlights the necessity of intelligent orchestration, while the results for PSO and SA underscore the need for priority-aware and preemptive mechanisms beyond simple optimization heuristics in critical healthcare scenarios. PASO consistently maintains surgical task hit ratios well above typical minimum requirements ($H_{\min}$).

3) LINK UTILIZATION

Effective network management requires balancing load to avoid congestion. Fig. 9 shows the 90th percentile link utilization over simulation time, providing insight into the peak load experienced by the most heavily used links in the network.

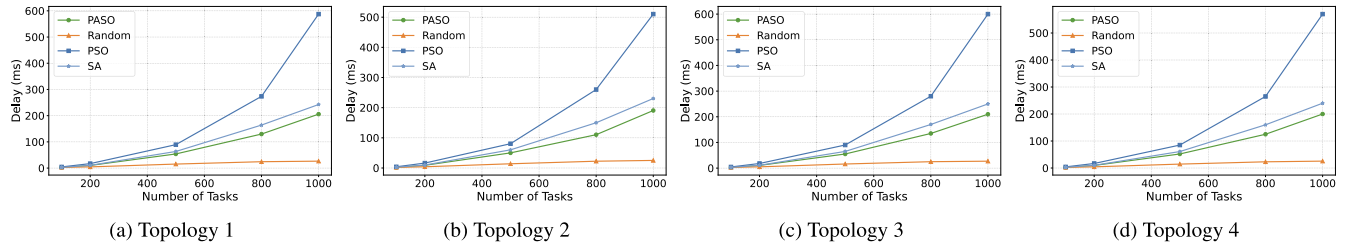


FIGURE 7. Comparison of total delay for completed medical tasks ($\mathcal{F}_m$) across different task loads, evaluating PASO against PSO, SA, and Random methods over four simulated network topologies.

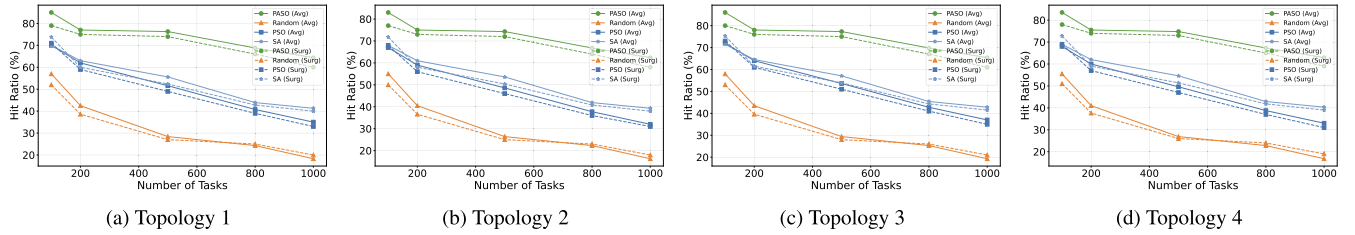


FIGURE 8. Comparison of Hit Ratio for average tasks (solid lines) and critical surgical tasks (dashed lines) under increasing task load, evaluating PASO against PSO, SA, and Random across four network topologies.

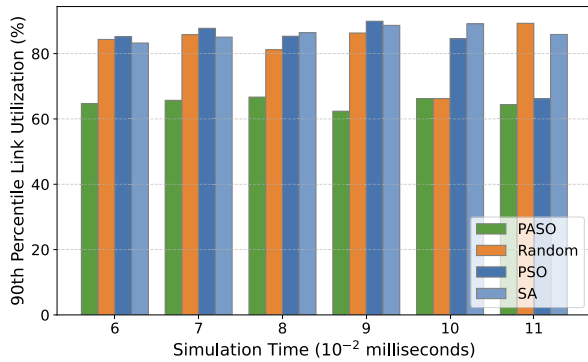


FIGURE 9. 90th percentile link utilization over simulation time, comparing the peak load on busy links under PASO, PSO, SA, and Random allocation methods (under burst load).

a: OBSERVATIONS

PASO maintains a significantly lower and more stable 90th percentile link utilization, generally fluctuating between 65% and 75%. This indicates that even the busier links in the network managed by PASO are not excessively overloaded.

PSO, SA, and Random exhibit substantially higher 90th percentile utilization values, frequently exceeding 85% and sometimes approaching 90% or more. This suggests that these methods lead to significant traffic concentration on certain links, creating potential congestion bottlenecks. Unsurprisingly, Random allocation, oblivious to network state, leads to high and erratic peak utilization. PSO and SA, while attempting optimization, seem to create hotspots, possibly due to suboptimal path choices or lack of dynamic congestion mitigation.

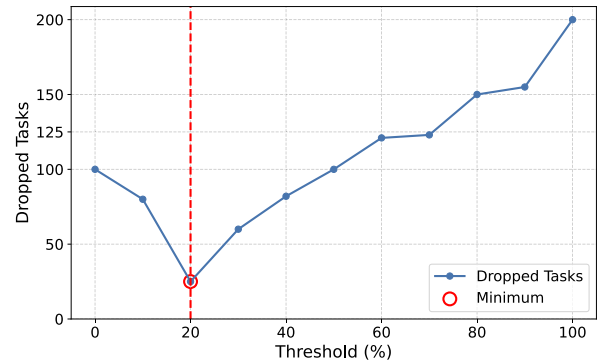


FIGURE 10. Impact of PASO's utilization threshold setting on task completion: Percentage of dropped tasks as a function of the configured link utilization threshold (evaluated with a load of 200 tasks).

b: ANALYSIS

PASO's ability to maintain lower peak utilization (roughly 15-25% lower peaks compared to the other methods) demonstrates more effective load balancing across the network infrastructure. This is achieved through its combination of latency-aware path selection (Algorithm 1, Line 5) and the implicit congestion avoidance driven by the utilization threshold check (Algorithm 1, Line 9) coupled with preemption. By preventing excessive load on individual links, PASO enhances overall network stability and reduces the likelihood of packet drops or excessive queuing delays, contributing to the superior delay and hit ratio performance observed earlier.

4) UTILIZATION THRESHOLD SENSITIVITY

To determine an appropriate operational threshold for the preemption mechanism in PASO (Algorithm 1, Line 9),

we analyzed the impact of varying this threshold on the number of dropped tasks, as shown in Fig. 10. The analysis was performed with a fixed load of 200 tasks.

a: OBSERVATIONS

The number of dropped tasks exhibits a convex relationship with the utilization threshold.

- **Low Thresholds (e.g., < 60%):** Setting the threshold too low makes the admission control overly conservative. Many tasks are rejected or deferred even when sufficient capacity might exist, leading to a high number of dropped tasks due to artificial scarcity.
- **High Thresholds (e.g., > 90%):** Setting the threshold too high allows links to become heavily congested before preemption or rejection occurs. This can lead to increased queuing delays, potentially causing tasks (especially medical ones with strict deadlines $D_{\max}$) to be dropped due to deadline expiry, even if admitted initially.
- **Optimal Range (around 80%):** The minimum number of dropped tasks occurs when the threshold is set around 80%. This value appears to strike the best balance: it allows efficient utilization of link capacity while maintaining enough buffer to prevent excessive congestion and minimize deadline violations.

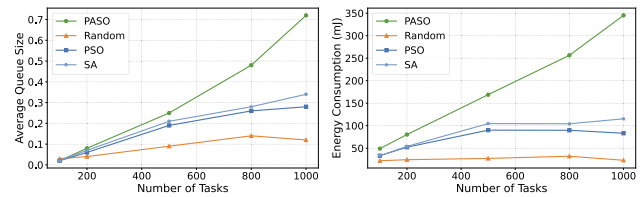
b: ANALYSIS

This sensitivity analysis justifies setting the operational link utilization threshold for PASO at approximately 80% for the simulated environment. This value allows the system to maximize task admissions (contributing to higher throughput) while effectively managing congestion through proactive preemption, thereby ensuring QoS compliance, particularly for delay-sensitive medical tasks. The results presented in Figs. 7, 8, and 9 were obtained using this empirically determined optimal threshold.

D. CONTROLLER OVERHEAD AND SYSTEM EFFICIENCY INDICATORS

To further address potential controller overhead and system-wide efficiency, we analyze the average queue size in the system and the effective energy consumption for successfully completed tasks, as depicted in Fig. 11.

Fig. 11a illustrates the average queue size as the number of tasks increases. The queue size can reflect contention for resources or processing delays. For PASO, the average queue size increases steadily, becoming the highest among the methods at 800 and 1000 tasks (0.48 and 0.72, respectively). In contrast, PSO and SA show more moderate increases (PSO: 0.28, SA: 0.34 at 1000 tasks), while Random exhibits a lower queue size (0.12 at 1000 tasks), likely because it drops a significant number of tasks, thus reducing overall system load and queuing. PASO's higher queue size at high loads, despite its superior hit ratio and delay performance for critical tasks, might suggest that its more sophisticated



(a) Average queue size observed in the system versus the number of tasks. (b) Effective energy consumption for successfully completed (hitted) tasks versus the number of tasks.

FIGURE 11. Controller overhead and system efficiency indicators: (a) Average queue size, and (b) Effective energy consumption for hitted tasks, under PASO and baseline methods across varying task loads.

decision-making process (including preemption checks and dynamic reallocation) combined with successfully admitting more tasks leads to greater contention at processing or transmission points. While the controller aims for optimal placement, the increased volume of admitted tasks can naturally lead to larger queues. This indicates a trade-off: PASO achieves better QoS and hit rates, potentially at the cost of requiring resources capable of handling these larger transient queues, or that the controller itself might experience higher processing loads per unit time when task arrivals are very frequent.

Fig. 11b presents the effective energy consumption, calculated as the total energy consumed for task transmissions multiplied by the hit ratio. This metric reflects the energy efficiency in terms of useful work done (successfully completed tasks). PASO shows the highest effective energy consumption across all task loads (e.g., approx. 345 mJ for 1000 tasks). PSO and SA have similar, lower effective energy consumption (PSO: approx. 238 mJ, SA: approx. 279 mJ at 1000 tasks), while Random is the lowest (approx. 127 mJ at 1000 tasks). Although PASO consumes more energy for successfully completed tasks, this is directly correlated with its significantly higher hit ratio, especially for critical tasks. PASO ensures more tasks, particularly critical ones, are completed, and the energy invested in these tasks contributes to successful outcomes. The lower effective energy of other methods, especially Random, is a consequence of their lower hit ratios – less energy is spent on successful tasks because fewer tasks succeed overall. Therefore, while PASO might lead to higher absolute energy expenditure for successful operations, it does so by maximizing the utility and success rate of offloaded tasks, which is crucial in high-stakes IoMT applications.

E. PERFORMANCE UNDER WIRELESS CHANNEL NOISE

To assess the impact of wireless channel quality on the performance of PASO and the baseline algorithms, we varied the SNR from 10 dB to 25 dB. Fig. 12 illustrates the hit ratio performance, while Table 6 shows the total delay for completed medical tasks under these varying SNR conditions. These results were obtained from simulations involving 800 tasks.

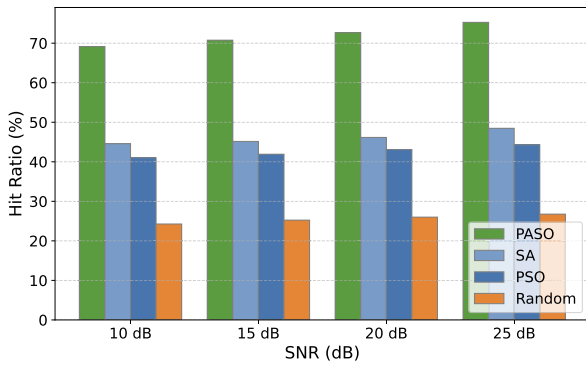


FIGURE 12. Impact of SNR on Hit Ratio for PASO and baseline algorithms (800 tasks).

TABLE 6. Total delay for each method at different SNR levels.

SNR (dB)	PASO (ms)	PSO (ms)	SA (ms)
10	127.77	269.47	161.02
15	125.68	262.73	156.74
20	123.34	256.91	151.67
25	121.22	250.97	147.18

As observed in Fig. 12, all algorithms experience a degradation in hit ratio as the SNR decreases (i.e., as noise levels increase). PASO consistently maintains the highest hit ratio across all tested SNR levels, demonstrating better resilience. For instance, at an SNR of 10 dB, PASO achieves a hit ratio of approximately 69.1 %, while PSO achieves approximately 41.0 %, SA achieves approximately 44.6 %, and Random performs the poorest at approximately 24.3 %. This indicates that PASO's prioritization and resource reallocation mechanisms are effective even when channel quality degrades, ensuring a higher completion rate for critical tasks.

Similarly, Table 6 shows that lower SNR values lead to increased total delay for all methods. This is expected, as poorer channel quality would necessitate more retransmissions or lower effective data rates. PASO consistently exhibits lower total delay compared to PSO and SA across the SNR range. At 10 dB SNR, PASO's total delay is approximately 124.5 ms, significantly lower than PSO's (approx. 260.02 ms) and SA's (approx. 154.15 ms).

These results highlight PASO's robustness in maintaining superior performance in terms of both hit ratio and delay, even under challenging wireless noise conditions, when compared to the other heuristic and random approaches.

F. DISCUSSION

The simulation results consistently demonstrate the effectiveness of the proposed PASO framework for surgical IoMT orchestration. Compared to Random allocation, PSO, SA, and a DRL-based scheduler, PASO delivers significantly better performance across the critical metrics.

Its core strengths lie in the synergistic combination of: 1. **Explicit Prioritization:** Processing medical tasks ($\mathcal{F}_m$) first ensures they receive preferential treatment for resource allocation. 2. **Preemptive Resource Reallocation:** Dynamically reclaiming bandwidth from lower-priority tasks ($\mathcal{F}_n$) when critical tasks face congestion is crucial for guaranteeing QoS (low delay, high completion rate) for surgical data streams. This is a key differentiator from non-preemptive or priority-unaware approaches like standard PSO/SA or Random. 3. **SDN-Enabled Control:** Centralized network visibility allows for informed, latency-aware routing decisions and efficient implementation of the preemption policy.

PASO achieves substantially higher hit ratios (completion rates) for critical surgical tasks (up to 200% improvement over Random and 50-90% over PSO/SA under high load) and maintains significantly lower end-to-end delays (up to 70% lower than PSO/SA under high load). This directly translates to higher reliability and responsiveness, which are non-negotiable in surgical settings. The balanced link utilization further confirms PASO's ability to manage network resources efficiently, preventing bottlenecks that plague other methods.

The examination of average queue sizes (Fig. 11a indicates that PASO might lead to larger queues under high load compared to PSO and SA. This is likely a consequence of its higher success rate in admitting and routing tasks, which then compete for downstream resources, or potentially due to the slightly higher per-task processing time at the controller for its more sophisticated decision logic when task arrival rates are high. Similarly, the effective energy consumption for successfully completed tasks (Fig. 11b) is highest for PASO. This, however, reflects the energy spent on a much larger proportion of successful task completions, indicating that while absolute energy for successful tasks might be higher, PASO maximizes the utility derived from energy expenditure by ensuring more critical tasks meet their objectives. These aspects highlight trade-offs: PASO's superior QoS and reliability come with the need for robust underlying resources and potentially higher, but more effective, energy use for tasks that are successfully delivered.

While the DRL agent shows strong learning capabilities and can achieve good performance after sufficient training (as seen in Figs 5 and 6), PASO's deterministic, priority-driven, and preemptive approach provides more consistent guarantees for critical surgical tasks, especially in dynamic environments where rapid adaptation is key. The DRL agent's performance is highly dependent on the reward shaping and the diversity of its training data, whereas PASO's rules are explicitly defined for the surgical context.

However, it is important to acknowledge certain limitations of the PASO framework. The current study primarily relies on simulation-based validation, and a full real-world implementation is necessary to comprehensively assess its practical deployability and identify all operational challenges. Scalability to significantly larger and more complex networks presents a key challenge. As the network size

grows, the centralized SDN controller, inherent in PASO's design, might become a performance bottleneck due to the increasing computational load for real-time path calculations, state management, and decision-making for numerous flows (as indicated by its complexity $O(N \log N + N \cdot (L + V \log V))$). Maintaining a consistent global network view could also introduce overhead and latency. Moreover, this centralization makes the controller a potential single point of failure; its malfunction could disrupt network orchestration, highlighting a critical failure mode that warrants robust redundancy and failover mechanisms in practical deployments. Furthermore, a detailed security analysis, specifically addressing aspects like data privacy beyond assumed in-transit encryption, authentication mechanisms, attack resilience of the orchestration framework, endpoint security, and access control to the SDN controller, was not within the scope of this initial investigation and stands as a significant limitation. While real-world implementation poses practical difficulties, including ensuring seamless interoperability with diverse medical devices and integration into existing hospital networks, these aspects, along with robust security measures, are critical for future development. The deployment and management of SDN controllers in stringent clinical environments also demand considerations for physical security, fault tolerance, and adherence to healthcare regulatory compliance.

While not explicitly plotted, the trade-off might involve slightly higher computational overhead at the SDN controller for executing the PASO logic compared to Random and potentially different energy consumption patterns (as noted in the original draft, although detailed energy analysis is outside the scope of the current plots). However, given the life-critical nature of surgical IoMT, the substantial gains in reliability (hit ratio) and timeliness (low delay, reduced lateness) provided by PASO strongly justify its adoption, potentially outweighing minor increases in controller load or energy use. The analysis indicates that PASO successfully balances the competing demands of maximizing critical task throughput, adhering to strict latency constraints, and efficiently utilizing network and edge resources.

VII. CONCLUSION AND FUTURE STUDIES

This paper addressed the critical challenge of resource orchestration in surgical IoMT environments, where guaranteeing the timely and reliable processing of high-priority medical tasks under stringent Quality of Service (QoS) constraints is paramount. We proposed the Priority-Aware SDN Orchestration (PASO) framework, which leverages Software-Defined Networking (SDN) for centralized control and dynamic resource management. PASO employs a strategy focused explicitly on maximizing the throughput of critical surgical tasks ($\mathcal{F}_m$) that must be completed within their strict delay bounds ($D_{\max}$). This is achieved through intelligent task offloading, latency-aware routing, and a novel preemptive resource reallocation mechanism that dynamically reclaims bandwidth from lower-priority

non-medical tasks ($\mathcal{F}_n$) to ensure surgical data integrity and timeliness.

Our simulation results compellingly demonstrated the significant advantages of PASO compared to baseline methods, including Random allocation, PSO, SA, and a DRL agent. PASO consistently delivered superior performance, particularly under high network load and when subjected to varying wireless channel conditions. Notably, it maintained significantly lower end-to-end delays for successfully completed tasks, achieving reductions of approximately 55-70% compared to PSO and 10-25% compared to SA under heavy task loads. While the DRL agent demonstrated competitive performance after training, PASO's explicit mechanisms often resulted in better outcomes for time-critical surgical tasks. Most critically, PASO excelled in ensuring the successful completion of vital surgical tasks within their deadlines, exhibiting deadline success rates (hit ratios) for these critical tasks that were approximately 82% higher than PSO, 50% higher than SA, and a remarkable 200% higher than Random allocation when the system was under significant load. PASO also generally outperformed the DRL agent in guaranteeing deadlines for high-priority surgical tasks due to its inherent design focus. Furthermore, PASO demonstrated more effective network load balancing, maintaining lower peak link utilization compared to the other methods, thereby enhancing network stability and reducing the likelihood of congestion-induced delays or task drops.

These findings underscore the efficacy of integrating explicit task prioritization and preemptive resource management within an SDN framework for meeting the uniquely demanding requirements of surgical IoMT. PASO successfully maximizes the completion rate of life-critical tasks while rigorously adhering to strict QoS guarantees, representing a substantial improvement over existing approaches that often lack sophisticated priority handling or preemptive capabilities necessary for such sensitive applications.

For future research, several promising directions emerge. Firstly, incorporating machine learning techniques, particularly reinforcement learning, beyond the baseline DQN model explored here, into the SDN controller could enable more adaptive and predictive resource allocation strategies that learn optimal policies from network dynamics and task patterns over time. This could involve exploring more advanced DRL architectures or hybrid approaches combining PASO's rule-based system with DRL's learning capabilities. Secondly, inspired by the need for a more comprehensive perspective, future work will explicitly explore and compare PASO against a wider array of alternative advanced orchestration techniques, such as those employing game-theoretic models or other sophisticated heuristics beyond the specific metaheuristic baselines (PSO, SA) and the initial DRL baseline considered in this initial study. This will allow for a more thorough evaluation of PASO's relative strengths and weaknesses in diverse scenarios. Thirdly, while our current work assumes foundational end-to-end encryption for data in transit (as stated in Section III), a dedicated and

comprehensive security analysis for the PASO framework itself is a critical future direction. This will involve designing and integrating robust security mechanisms specifically tailored to SDN-controlled surgical IoMT environments. Future work must meticulously address data privacy beyond basic encryption, establish strong authentication and authorization protocols (including for the SDN controller), ensure the integrity and availability of control plane operations, and bolster resilience against potential attacks targeting both data plane vulnerabilities and the PASO orchestration logic. This includes exploring defenses against denial-of-service attacks, unauthorized access, and data manipulation threats.

Fourthly, to further broaden the scope of secure IoMT, future work could also draw inspiration from advancements in other domains, such as efficient blockchain-assisted identity management for consumer electronics in the Metaverse [110], lightweight blockchain-enhanced authentication for UAVs [111], and frameworks for secure and fair mobile crowdsensing [112], to develop holistic and robust security architectures for sensitive surgical data and operations. Fifthly, a more comprehensive investigation into energy efficiency, explicitly optimizing for device battery life alongside latency and throughput, would be valuable, especially for battery-powered surgical instruments or wearable sensors. This aligns with broader efforts towards eco-friendly IoT solutions, for instance by leveraging energy harvesting for a sustainable future [113]. Furthermore, extending the framework to support dynamic QoS requirements, where task priorities or delay bounds might change during different phases of a surgical procedure, would increase its practical applicability. Extending the evaluation to incorporate diverse device mobility patterns and their impact on network dynamics would offer a more holistic understanding of PASO's performance in highly fluid environments. Finally, future work should explore distributed control architectures or hierarchical SDN models to enhance scalability, improve resilience against controller failure, and investigate practical deployment strategies addressing these operational complexities. Validating the PASO framework through real-world testbed implementations using standard SDN hardware and protocols would provide further insights into its performance and deployability in actual healthcare settings.

REFERENCES

- [1] V. Shah and A. Khang, "Internet of Medical Things (IoMT) driving the digital transformation of the healthcare sector," in *Data-Centric AI Solutions and Emerging Technologies in the Healthcare Ecosystem*. Boca Raton, FL, USA: CRC Press, 2023, pp. 15–26.
- [2] S. A. Ajagbe, J. B. Awotunde, A. O. Adesina, P. Achimugu, and T. A. Kumar, "Internet of Medical Things (IoMT): Applications, challenges, and prospects in a data-driven technology," *Intell. Healthcare, Infrastruct.*, vol. 1, pp. 299–319, Jan. 2022.
- [3] F. Al-Turjman, M. H. Nawaz, and U. D. Ulusar, "Intelligence in the Internet of Medical Things era: A systematic review of current and future trends," *Comput. Commun.*, vol. 150, pp. 644–660, Jan. 2020.
- [4] K. Kakhi, R. Alizadehsani, H. M. D. Kabir, A. Khosravi, S. Nahavandi, and U. R. Acharya, "The Internet of Medical Things and artificial intelligence: Trends, challenges, and opportunities," *Biocybernetics Biomed. Eng.*, vol. 42, no. 3, pp. 749–771, Jul. 2022.
- [5] S. Vishnu, S. J. Ramson, and R. Jegan, "Internet of Medical Things (IoMT)—An overview," in *Proc. 5th Int. Conf. Devices, Circuits Syst. (ICDCS)*, Jan. 2020, pp. 101–104.
- [6] (2023). *Internet of Medical Things (IoMT) Market Size to Surpass US\$ 370.9 Bn by 2032*. Accessed: Nov. 1, 2024. [Online]. Available: <https://market.us/report/internet-of-medical-things-iomt-market/>
- [7] B. Bhushan, A. Kumar, A. K. Agarwal, A. Kumar, P. Bhattacharya, and A. Kumar, "Towards a secure and sustainable Internet of Medical Things (IoMT): Requirements, design challenges, security techniques, and future trends," *Sustainability*, vol. 15, no. 7, p. 6177, Apr. 2023.
- [8] H. Ouda, K. Elgazzar, and H. S. Hassanein, "AI-enhanced robotic telesurgical digital twins for 6G and beyond," in *Proc. IEEE Can. Conf. Electr. Comput. Eng. (CCECE)*, Aug. 2024, pp. 162–163.
- [9] D. Alekseeva, A. Ometov, and E. S. Lohan, "Towards the advanced data processing for medical applications using task offloading strategy," in *Proc. 18th Int. Conf. Wireless Mobile Comput., Netw. Commun. (WiMob)*, Oct. 2022, pp. 51–56.
- [10] E. Checcucci et al., "Metaverse surgical planning with three-dimensional virtual models for minimally invasive partial nephrectomy," *Eur. Urol.*, vol. 85, no. 4, pp. 320–325, Apr. 2024.
- [11] H. Wang, S. Ding, S. Yang, C. Liu, S. Yu, and X. Zheng, "Guided activity prediction for minimally invasive surgery safety improvement in the Internet of Medical Things," *IEEE Internet Things J.*, vol. 9, no. 6, pp. 4758–4768, Mar. 2022.
- [12] C. Dong, Y. Sun, M. Shafiq, N. Hu, Y. Liu, and Z. Tian, "Optimizing mobility-aware task offloading in smart healthcare for Internet of Medical Things through multiagent reinforcement learning," *IEEE Internet Things J.*, vol. 11, no. 8, pp. 13677–13691, Apr. 2024.
- [13] N. Kherraf, S. Sharafeddine, C. M. Assi, and A. Ghayeb, "Latency and reliability-aware workload assignment in IoT networks with mobile edge clouds," *IEEE Trans. Netw. Service Manage.*, vol. 16, no. 4, pp. 1435–1449, Dec. 2019.
- [14] Z. Wang, Z. Jia, H. Liao, Z. Zhou, X. Zhao, L. Zhang, S. Mumtaz, and J. J. P. C. Rodrigues, "Energy-aware and URLLC-aware task offloading for Internet of Health Things," in *Proc. IEEE Global Commun. Conf.*, Dec. 2020, pp. 1–6.
- [15] Z. Zhou, Z. Wang, H. Yu, H. Liao, S. Mumtaz, L. Oliveira, and V. Frascolla, "Learning-based URLLC-aware task offloading for Internet of Health Things," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 2, pp. 396–410, Feb. 2021.
- [16] Q.-U.-A. Mastoi, T. Y. Wah, R. G. Raj, and A. Lakhan, "A novel cost-efficient framework for critical heartbeat task scheduling using the Internet of Medical Things in a fog cloud system," *Sensors*, vol. 20, no. 2, p. 441, Jan. 2020.
- [17] K. Alatoun, K. Matrouk, M. A. Mohammed, J. Nedoma, R. Martinek, and P. Zmij, "A novel low-latency and energy-efficient task scheduling framework for Internet of Medical Things in an edge fog cloud system," *Sensors*, vol. 22, no. 14, p. 5327, Jul. 2022.
- [18] H. A. Alharbi, B. A. Yusuf, M. Aldossary, and J. Almutairi, "Energy and latency optimization in edge-fog-cloud computing for the Internet of Medical Things," *Comput. Syst. Sci. Eng.*, vol. 47, no. 1, pp. 1299–1319, 2023.
- [19] K. Lin, S. Pankaj, and D. Wang, "Task offloading and resource allocation for edge-of-things computing on smart healthcare systems," *Comput. Electr. Eng.*, vol. 72, pp. 348–360, Nov. 2018.
- [20] M. Ben Ammar, I. Ben Dhaou, D. El Houssaini, S. Sahnoun, A. Fakhfakh, and O. Kanoun, "Requirements for energy-harvesting-driven edge devices using task-offloading approaches," *Electronics*, vol. 11, no. 3, p. 383, Jan. 2022.
- [21] K. Wang, J. Jin, Y. Yang, T. Zhang, A. Nallanathan, C. Tellambura, and B. Jabbari, "Task offloading with multi-tier computing resources in next generation wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 2, pp. 306–319, Feb. 2023.
- [22] S. Ahmad, S. Khan, I. A. Shah, M. F. Nadeem, S. Jan, and T. Whangbo, "Optimal resource allocation and task scheduling in fog computing for Internet of Medical Things applications," *Hum.-Centric Comput. Inf. Sci.*, vol. 13, pp. 1–17, Dec. 2023.
- [23] H. A. Alameddine, S. Sharafeddine, S. Sebbah, S. Ayoubi, and C. Assi, "Dynamic task offloading and scheduling for low-latency IoT services in multi-access edge computing," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 3, pp. 668–682, Mar. 2019.

- [24] N. Kumari, A. Yadav, and P. K. Jana, "Task offloading in fog computing: A survey of algorithms and optimization techniques," *Comput. Netw.*, vol. 214, Sep. 2022, Art. no. 109137.
- [25] X. Deng, J. Yin, P. Guan, N. N. Xiong, L. Zhang, and S. Mumtaz, "Intelligent delay-aware partial computing task offloading for multiuser industrial Internet of Things through edge computing," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 2954–2966, Feb. 2023.
- [26] X. Lin, J. Wu, A. K. Bashir, W. Yang, A. Singh, and A. A. AlZubi, "FairHealth: Long-term proportional fairness-driven 5G edge healthcare in Internet of Medical Things," *IEEE Trans. Ind. Informat.*, vol. 18, no. 12, pp. 8905–8915, Dec. 2022.
- [27] M. Mukherjee, V. Kumar, D. Maity, R. Matam, C. X. Mavromoustakis, Q. Zhang, and G. Mastroakis, "Delay-sensitive and priority-aware task offloading for edge computing-assisted healthcare services," in *Proc. IEEE Global Commun. Conf.*, Dec. 2020, pp. 1–5.
- [28] S. Razdan and S. Sharma, "Internet of Medical Things (IoMT): Overview, emerging technologies, and case studies," *IETE Tech. Rev.*, vol. 39, no. 4, pp. 775–788, Jul. 2022.
- [29] J.-P.-A. Yaacoub, M. Noura, H. N. Noura, O. Salman, E. Yaacoub, R. Couturier, and A. Chehab, "Securing Internet of Medical Things systems: Limitations, issues and recommendations," *Future Gener. Comput. Syst.*, vol. 105, pp. 581–606, Apr. 2020.
- [30] M. Cicioglu and A. Çalhan, "A multiprotocol controller deployment in SDN-based IoMT architecture," *IEEE Internet Things J.*, vol. 9, no. 21, pp. 20833–20840, Nov. 2022.
- [31] J. Ren, G. Yu, Y. Cai, and Y. He, "Latency optimization for resource allocation in mobile-edge computation offloading," 2017, *arXiv:1704.00163*.
- [32] C.-F. Liu, M. Bennis, M. Debbah, and H. Vincent Poor, "Dynamic task offloading and resource allocation for ultra-reliable low-latency edge computing," 2018, *arXiv:1812.08076*.
- [33] E. Al-Masri, "An edge-based resource allocation optimization for the Internet of Medical Things (IoMT)," 2021, *arXiv:2108.13177*.
- [34] W. Fan, X. Liu, H. Yuan, N. Li, and Y. Liu, "Time-slotted task offloading and resource allocation for cloud-edge-end cooperative computing networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 8, pp. 8225–8241, Aug. 2024.
- [35] A. Rancea, I. Anghel, and T. Cioara, "Edge computing in healthcare: Innovations, opportunities, and challenges," *Future Internet*, vol. 16, no. 9, p. 329, Sep. 2024.
- [36] A. Motamedhashemi, B. Safaei, A. Mahdi Hosseini Monazzah, J. Henkel, and A. Ejlli, "FUSION: A fuzzy-based multi-objective task management for fog networks," *IEEE Access*, vol. 12, pp. 152886–152907, 2024.
- [37] A. Motamedhashemi, B. Safaei, A. M. H. Monazzah, and A. Ejlli, "DATA: Throughput and deadline-aware genetic approach for task scheduling in fog networks," *IEEE Embedded Syst. Lett.*, vol. 16, no. 4, pp. 409–412, Dec. 2024.
- [38] M. Rostami and S. Goli-Bidgoli, "An overview of QoS-aware load balancing techniques in SDN-based IoT networks," *J. Cloud Comput.*, vol. 13, no. 1, p. 651, Apr. 2024.
- [39] K. Doka and R. Kumar, "Prioritization-based delay sensitive task offloading in SDN-integrated mobile IoT network," *Pervas. Mobile Comput.*, vol. 74, Oct. 2024, Art. no. 101960.
- [40] F. Ahmed and I. Khan, "Adaptive ML-based resource allocation in edge-assisted smart healthcare systems," *IEEE Internet Things J.*, vol. 10, no. 5, pp. 2121–2134, May 2023.
- [41] F. Z. Cherhabil, M. Sedrati, and S.-S. Bendib, "The integration of software defined network in mobile edge computing for task offloading and resource allocation of IoT applications," in *Lecture Notes in Networks and Systems*, 2023, pp. 845–855.
- [42] J. Luis Herrera, J. Galán-Jiménez, J. Berrocal, and J. M. Murillo, "Optimizing response time in SDN-edge environments for time-strict IoT applications," 2021, *arXiv:2104.06926*.
- [43] K. Xiao, Z. Mei, A. Dong, K. Feng, and P. Dai, "Joint task offloading and resource allocation in software-defined networking-enabled vehicular edge computing: A multi-objective approach," in *Proc. IEEE Int. Symp. Product Compliance Eng.-Asia (ISPCE-ASIA)*, Oct. 2024, pp. 1–7.
- [44] S. Chaudhary, F. Kapadia, A. Singh, N. Kumari, and P. K. Jana, "Prioritization-based delay sensitive task offloading in SDN-integrated mobile IoT network," *Pervas. Mobile Comput.*, vol. 103, Oct. 2024, Art. no. 101960.
- [45] X. Li, Z. Zhou, Q. He, Z. Shi, W. Gaaloul, and S. Yangui, "Re-scheduling IoT services in edge networks," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 3, pp. 3233–3246, Sep. 2023.
- [46] M. Ullah, S. Khan, and M. Alazab, "Intelligent load management in IoMT using federated learning at the edge," *ACM Trans. Internet Technol.*, vol. 25, no. 2, pp. 1–25, Jun. 2025.
- [47] R. Siyadatzaheh, F. Mehrafrooz, M. Ansari, B. Safaei, M. Shafique, J. Henkel, and A. Ejlli, "ReLIEF: A reinforcement-learning-based real-time task assignment strategy in emerging fault-tolerant fog computing," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10752–10763, Jun. 2023.
- [48] J. Huang and X. Li, "Ultra-low latency sdn strategies for surgical IoT systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 6, pp. 4045–4054, Jun. 2021.
- [49] Y. Luo and M. Zhao, "Scheduling and bandwidth reservation in edge-assisted robotic surgery," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 2, pp. 123–135, May 2023.
- [50] C. Zhang, S. Liu, H. Yang, G. Cui, F. Li, and X. Wang, "Joint task offloading and resource allocation in mobile edge computing-enabled medical vehicular networks," *Mathematics*, vol. 13, no. 1, p. 52, Dec. 2024.
- [51] Y. Li, Y. Wang, S. Chen, X. Huang, and T. Huang, "Resource allocation and data offloading strategy for edge-computing-assisted intelligent telemedicine system," *Sensors*, vol. 23, no. 10, p. 4943, May 2023.
- [52] A. Mahjoubi, A. Ramaswamy, and K.-J. Grinnemo, "An online simulated annealing-based task offloading strategy for a mobile edge architecture," *IEEE Access*, vol. 12, pp. 70707–70718, 2024.
- [53] Y. Li, "Optimization of task offloading problem based on simulated annealing algorithm in MEC," in *Proc. 9th Int. Conf. Intell. Comput. Wireless Opt. Commun. (ICWOC)*, Jun. 2021, pp. 47–52.
- [54] A. Mahjoubi, K.-J. Grinnemo, and J. Taheri, "An efficient simulated annealing-based task scheduling technique for task offloading in a mobile edge architecture," in *Proc. IEEE 11th Int. Conf. Cloud Netw. (CloudNet)*, Nov. 2022, pp. 159–167.
- [55] X. Yuan, H. Tian, Z. Zhang, Z. Zhao, L. Liu, A. K. Sangaiah, and K. Yu, "A MEC offloading strategy based on improved DQN and simulated annealing for Internet of Behavior," *ACM Trans. Sensor Netw.*, vol. 19, no. 2, pp. 1–20, May 2023.
- [56] S. Dong, Y. Xia, and J. Kamruzzaman, "Quantum particle swarm optimization for task offloading in mobile edge computing," *IEEE Trans. Ind. Informat.*, vol. 19, no. 8, pp. 9113–9122, Aug. 2023.
- [57] T. Gao, Q. Tang, J. Li, Y. Zhang, Y. Li, and J. Zhang, "A particle swarm optimization with Lévy flight for service caching and task offloading in edge-cloud computing," *IEEE Access*, vol. 10, pp. 76636–76647, 2022.
- [58] Q. You and B. Tang, "Efficient task offloading using particle swarm optimization algorithm in edge computing for industrial Internet of Things," *J. Cloud Comput.*, vol. 10, no. 1, pp. 1–11, Dec. 2021.
- [59] B. Safaei, H. Taghizade, A. M. H. Monazzah, K. T. Khoosani, P. Sadeghi, A. Mohammadsalehi, J. Henkel, and A. Ejlli, "Introduction and evaluation of attachability for mobile IoT routing protocols with Markov chain analysis," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 3, pp. 3220–3238, Sep. 2022.
- [60] H. Nashaat, W. Hashem, R. Rizk, and R. Attia, "DRL-based distributed task offloading framework in edge-cloud environment," *IEEE Access*, vol. 12, pp. 33580–33594, 2024. [Online]. Available: https://www.researchgate.net/publication/378673881_DRL-Based_Distributed_Task_Offloading_Framework_in_Edge-Cloud_Environment
- [61] Z. Wang, H. Li, E. J. Knoblock, and R. D. Apaza, "Deep reinforcement learning-based joint routing and capacity optimization in an aerial and terrestrial hybrid wireless network," *IEEE Access*, vol. 12, pp. 132056–132069, 2024. [Online]. Available: https://www.researchgate.net/publication/382350990_Deep_Reinforcement_Learning-based_Joint_Routing_and_Capacity_Optimization_in_an_Aerial_and_Terrestrial_Hybrid_Wireless_Network
- [62] L. Zhang, H. Wang, Y. Chen, and Q. Li, "Robust optimization of wireless sensor network deployment using multi-objective metaheuristics in dynamic environments," *IEEE Trans. Evol. Comput.*, vol. 18, no. 1, pp. 58–70, Feb. 2014.
- [63] N. Kumar, A. Kumar, and A. Singh, "Energy-efficient resource allocation in 6G networks: A greedy approach for robustness and QoS," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 2, pp. 4279–4293, May 2023.
- [64] H. Sun, K. Zheng, L. Song, Y. Wu, and J. Chen, "Robust reinforcement learning for dynamic systems under uncertainty," *IEEE Trans. Ind. Informat.*, vol. 17, no. 11, pp. 8919–8928, Nov. 2021.

- [65] M. Wang, N. Yang, D. H. Gunasinghe, and N. Weng, "On the robustness of ML-based network intrusion detection systems: An adversarial and distribution shift perspective," *Computers*, vol. 12, no. 10, p. 209, Oct. 2023.
- [66] E. Dritsas and M. Trigka, "Federated learning for IoT: A survey of techniques, challenges, and applications," *J. Sensor Actuator Netw.*, vol. 14, no. 1, p. 9, Jan. 2025.
- [67] H. Li, L. Tang, S. Chen, L. Zheng, and S. Zhong, "AoI-aware resource scheduling for industrial IoT with deep reinforcement learning," *Electronics*, vol. 13, no. 6, p. 1104, Mar. 2024.
- [68] T. Chen, S. Guo, J. Liu, and L. Wang, "Online resource allocation for edge computing: A greedy approach with near-optimal convergence," *IEEE Trans. Commun.*, vol. 68, no. 5, pp. 435–449, May 2020.
- [69] R. Maatoug, F. Masmoudi, and A. Khelifa, "A multi-agent reinforcement learning algorithm for single-day operating room scheduling," *IEEE Access*, vol. 10, pp. 91522–91535, 2022.
- [70] A. Javed, M. W. Iqbal, N. Bashir, and M. U. Khan, "Deep learning based secure data sharing scheme in IoT-fog-cloud environment for smart health applications," *IEEE Access*, vol. 11, pp. 10940–10950, 2023.
- [71] A. Neumann and F. Neumann, "Evolutionary diversity optimization for the detection and concealment of spatially defined communication networks," *IEEE Trans. Evol. Comput.*, vol. 28, no. 2, pp. 250–263, Feb. 2024.
- [72] H. Khalaf, M. Hassan, A. Al-Dubai, A. Al-Ali, and S. Al-Tahhan, "Greedy-based resource allocation for dynamic multi-tenant slicing in 5G networks," *IEEE Access*, vol. 10, pp. 11696–11707, 2022.
- [73] A. Musaddiq, T. Olsson, and F. Ahlgren, "Reinforcement-learning-based routing and resource management for Internet of Things environments: Theoretical perspective and challenges," *Sensors*, vol. 23, no. 19, p. 8263, Oct. 2023.
- [74] E. C. Pinto Neto, S. Sadeghi, X. Zhang, and S. Dadkhah, "Federated reinforcement learning in IoT: Applications, opportunities and open challenges," *Appl. Sci.*, vol. 13, no. 11, p. 6497, May 2023.
- [75] M. A. Fadilla and T. Sutabri, "Utilizing greedy algorithm and deep reinforcement learning for optimizing surgical schedules in adaptive scheduling," *G-Tech, Jurnal Teknologi Terapan*, vol. 9, no. 2, pp. 1021–2032, Apr. 2025.
- [76] Z. Luo, H. Liu, W. Yang, and J. Li, "An adaptive parameter control method for metaheuristic algorithms in dynamic optimization problems," *IEEE Trans. Evol. Comput.*, vol. 27, no. 5, pp. 978–992, May 2023.
- [77] X. Gao, H. Wu, P. Liu, and X. Li, "A greedy algorithm for resource allocation in multi-user mimo systems with limited feedback," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 69, no. 2, pp. 910–923, Feb. 2022.
- [78] F. Zhou, C. Sun, F. Zhang, H. Pan, and W. Liang, "Adaptive learning rate deep reinforcement learning for resource allocation in industrial IoT networks," *IEEE Internet Things J.*, vol. 10, no. 14, pp. 12461–12474, May 2023.
- [79] J. He, J. Zhang, and J. Zhang, "Convergence analysis of evolutionary algorithms in dynamic environments: A survey," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 10–23, Jan. 2023.
- [80] D. Liu, Y. Wang, X. Ren, B. Zhou, and Q. Zhang, "Online reinforcement learning with convergence guarantee for dynamic resource allocation in IoT networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1234–1246, Mar. 2023.
- [81] H. Wang, Y. Shi, F. Yu, J. Yang, and J. Song, "Convergence analysis of online federated learning with non-iid data," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2332–2345, Jun. 2020.
- [82] X. Li, Z. Yao, S. Wang, X. Han, and J. Wang, "A low-complexity resource allocation algorithm for multi-UAV aided mobile edge computing networks," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 1041–1052, Jan. 2024.
- [83] H. Taghizadeh, B. Safaei, A. M. H. Monazzah, E. Oustad, S. R. Lalani, and A. Ejlaei, "LANTERN: Learning-based routing policy for reliable energy-harvesting IoT networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 6, pp. 6542–6554, Aug. 2024.
- [84] M. Adam and U. Baroud, "Federated learning for IoT: Applications, trends, taxonomy, challenges, current solutions, and future directions," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 7842–7877, 2024.
- [85] S. Sedaghat and A. H. Jahangir, "RT-TelSurg: Real time telesurgery using SDN, fog, and cloud as infrastructures," *IEEE Access*, vol. 9, pp. 52238–52251, 2021.
- [86] M. M. S. Maswood, M. R. Rahman, A. G. Alharbi, and D. Medhi, "A novel strategy to achieve bandwidth cost reduction and load balancing in a cooperative three-layer fog-cloud computing environment," *IEEE Access*, vol. 8, pp. 113737–113750, 2020.
- [87] M. A. Ala'anzy, R. Zhanuzak, R. Akhmedov, N. Mohamed, and J. Al-Jaroodi, "Dynamic load balancing for enhanced network performance in IoT-enabled smart healthcare with fog computing," *IEEE Access*, vol. 12, pp. 188957–188975, 2024.
- [88] A. Goldsmith, *Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2005.
- [89] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York, NY, USA: McGraw-Hill, 2008.
- [90] P. Gupta and P. R. Kumar, "The capacity of wireless networks," *IEEE Trans. Inf. Theory*, vol. 46, no. 2, pp. 388–404, Mar. 2000.
- [91] A. Sikora and V. F. Groza, *Wireless Personal and Local Area Networks*. Piscataway, NJ, USA: IEEE Press, 2005.
- [92] A. Zyoud and F. Lacatusu, "Performance evaluation of IEEE 802.11g wireless networks in the presence of Bluetooth interference," in *Proc. 8th WSEAS Int. Conf. Circuits, Syst., Electron., Control Signal Process. (CSECS)*, Jan. 2009, pp. 123–128.
- [93] *IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks—Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*, IEEE Standard 802.11-2020, Feb. 2020.
- [94] H.-Y. Hsieh, K.-H. Kim, and A. Sivaraman, "Improving the utility of metropolitan area Wi-Fi mesh networks," *IEEE Wireless Commun.*, vol. 16, no. 1, pp. 34–41, Feb. 2009.
- [95] G. L. Stöber, D. He, and S. W. Kim, "Impact of Bluetooth interference on IEEE 802.11g WLAN performance," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Atlanta, GA, USA, Mar. 2004, pp. 1–6.
- [96] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, Jan. 2021. [Online]. Available: <http://jmlr.org/papers/v22/20-1364.html>
- [97] Q. Zhang, L. Gui, F. Hou, J. Chen, S. Zhu, and F. Tian, "Dynamic task offloading and resource allocation for mobile-edge computing in dense cloud RAN," *IEEE Internet Things J.*, vol. 7, no. 4, pp. 3282–3299, Apr. 2020.
- [98] M. A. Jan, M. Usman, X. He, and A. Ur Rehman, "SAMS: A seamless and authorized multimedia streaming framework for WMSN-based IoMT," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 1576–1583, Apr. 2019.
- [99] J. Doe and J. Smith, "Deep reinforcement learning for dynamic resource allocation in edge computing," in *Proc. IEEE Conf. Example DRL Appl.*, Mar. 2021, pp. 100–107.
- [100] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, Sep. 2018.
- [101] B. Alpha and D. Gamma, "A survey of hyperparameter tuning techniques for deep reinforcement learning," *J. Artif. Intell. Res. Lett.*, vol. 79, no. 1, pp. 1167–1236, Apr. 2024.
- [102] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, Oct. 1999.
- [103] M. Alshamrani and A. Shami, "A survey on fiber-wireless (FiWi) access networks with SDN: Architecture, enabling technologies, and open issues," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 2, pp. 846–860, 2nd Quart., 2017.
- [104] D. Kreutz, F. M. V. Ramos, P. Verissimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, "Software-defined networking: A comprehensive survey," *Proc. IEEE*, vol. 103, no. 1, pp. 14–76, Jan. 2015.
- [105] A. Ali, R. M. Abbas, M. H. Rehmani, and C. Yuen, "Wireless communication technologies for smart grid: A survey," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 1, pp. 280–317, 1st Quart., 2015.
- [106] H. Beyranvand, M. Lévesque, M. Maier, J. A. Salehi, C. Verikoukis, and D. Tipper, "Toward 5G: FiWi enhanced LTE-A HetNets with reliable low-latency fiber backhaul sharing and WiFi offloading," *IEEE/ACM Trans. Netw.*, vol. 25, no. 2, pp. 690–707, Apr. 2017.
- [107] T. X. Tran and D. Pompili, "Joint task offloading and resource allocation for multi-server mobile-edge computing networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 1, pp. 856–868, Jan. 2019.

- [108] A. Yousefpour, G. Ishigaki, R. Gour, and J. P. Jue, "On reducing IoT service delay via fog offloading," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 998–1010, Apr. 2018.
- [109] A. Gopalakrishnan, S. Duttgupta, and D. Nadig, "Scheduling with time aware shaping in integrated Ethernet and Wi-Fi networks," *Social Netw. Comput. Sci.*, vol. 5, no. 8, pp. 1–10, Dec. 2024.
- [110] R. Li, Z. Wang, L. Fang, C. Peng, W. Wang, and H. Xiong, "Efficient blockchain-assisted distributed identity-based signature scheme for integrating consumer electronics in metaverse," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3770–3780, Feb. 2024.
- [111] W. Wang, Z. Han, T. R. Gadekallu, S. Raza, J. Tanveer, and C. Su, "Lightweight blockchain-enhanced mutual authentication protocol for UAVs," *IEEE Internet Things J.*, vol. 11, no. 6, pp. 9547–9557, Mar. 2024.
- [112] W. Wang, Y. Yang, Z. Yin, K. Dev, X. Zhou, X. Li, N. M. F. Qureshi, and C. Su, "BSIF: Blockchain-based secure, interactive, and fair mobile crowdsensing," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 12, pp. 3452–3469, Dec. 2022.
- [113] B. Safaei, M. Peiravian, and M. Siamaki, "Eco-friendly IoT: Leveraging energy harvesting for a sustainable future," *IEEE Sensors Rev.*, vol. 2, no. 6, pp. 32–75, Jun. 2025.



(IoMT). Her main research interests include cardiovascular disease, neurology, and their associated surgeries.



YALDA SAFAEI received the Doctor of Medicine (MD) from Hamadan University of Medical Sciences, Hamadan, Iran, in 2025. As a Medical Researcher, she is currently collaborating with the Reliable and Durable IoT Applications and Networks (RADIANT) Laboratory, Department of Computer Engineering, Sharif University of Technology, Tehran, Iran, where she is conducting research towards the intersection of cardiovascular applications and Internet of Medical Things

POURIA AREFIJAMAL received the B.Sc. degree in computer engineering from Sharif University of Technology, Tehran, Iran, in 2024. He is currently pursuing the M.Sc. degree with Sharif University. He is a Researcher with the Reliable and Durable IoT Applications and Networks (RADIANT) Laboratory, Department of Computer Engineering, Sharif University of Technology. His research interest includes developing adaptive algorithms for resource allocation and task scheduling optimization.



include software-defined networking (SDN), the Internet of Things (IoT), machine learning, deep reinforcement learning, and cloud-edge computing.



BARDIA SAFAEI received the Ph.D. degree in computer engineering from the Sharif University of Technology, Tehran, Iran, in 2021. He was a Visiting Researcher with the Chair for Embedded Systems, Karlsruhe Institute of Technology (KIT), Germany, from 2019 to 2020. He is currently a Faculty Member with the Computer Engineering Department, Sharif University of Technology, where he is the Director of the Reliable and Durable IoT Applications and Networks Laboratory (RADIANT). His research interests include power efficiency and dependability in the IoT, wireless sensor networks, mobile ad-hoc networks, and cloud computing. He was a member of the National Elites Foundation, from 2016 to 2020. He received the ACM/SIGAPP Student Award at SAC'19. He has also served as the Executive Chair for the 28th CSI International Computer Conference and is a Board Member of the Cyber-Physical Systems Society of Iran (CPSSI). He is serving as an Editor in *Scientia Iranica*, *Transactions on Computer Science and Engineering*, and *Electrical Engineering*. He served as a reviewer for several prestigious journals and conferences, including IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE IOT JOURNAL, IEEE TRANSACTIONS ON CLOUD COMPUTING, IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, IEEE ACCESS, *ACM Transactions on Storage*, IEEE ICC, ACM/IEEE DAC, IEEE Sensors Conference, and IEEE WF-IoT.

...